

Code explanation (Call Nr. 8)

2D Fokker–Planck solver, showing advantages and limits of the novel diagonal solver

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Comment on the document

This document describes mostly mathematical backgrounds of what we worked on in the last weeks. Ultimately the whole reasoning aims to explaining why our proposed scheme is indeed better than the naive stencil implementation. Some parts in this document are a bit verbose, because we will maybe use some parts in our final report.

It turned out to be very hard to find good examples where the naive stencil scheme breaks down, while using physical values for our D-tensor. It is not hard to find examples just using absurdly high values for D_{xy} , but a construction that uses physically meaningful D_{xy} is very challenging.

In Section 1 we explain why we later use certain positivity assumptions on D_{xy} , D_{xx} , and D_{yy} . Section 2 summarizes again why we need positivity on $\tilde{A}_x, \tilde{A}_y, \tilde{D}_2$.

Last call we discussed bounds on our mesh-size-ratio r . You proposed that we find a general bound for this. Section 3 does the full derivation on a general bound which holds for any kind of model.

Section 4 shows us the real problem of the naive 9-point implementation. The scheme itself works totally fine, expect that it does not preserve positivity. Besides that it is stable (under our physical D-assumptions). It is based on the simple methodology we did a few weeks ago on a call with Mr. Monasse.

Section 5 does a Von Neumann Analysis of the diagonal method. This formalizes what we did in Section 4 a bit more and gives us the concrete CFL bounds we are using in the code. I felt a proper derivation of these bounds was missing so far.

Section 6 goes into more detail about the implementation of how a non-uniform grid could look like. During these maths we uncovered an important limitation of the proposed scheme, which is presented there.

Experiments to numerically show why our scheme works better than the naive 9-point stencil are shown in Section 7. This was particularly challenging and probably presents the most important part of this document (wrapping up all we did before mathematically).

To show how our scheme behaves on longer runs, we show various experiments in Section 8. It is interesting to note that with our changing concentrations C_{11} and C_{10} our distributions converge after little time. After this the system stays unchanged and the adaptive timestepping chooses large timesteps. We are not sure if this is only the case in our simplified model.

1 Physical setup and the Fokker–Planck equation

In this section I tried to formalize positivity properties of our Diffusion tensor, because my recent documents never worked this out explicitly.

Everything which is orange in this document is a repetition of the last week’s document and only here for reference.

1.1 Master equation, drift and diffusion

We model the density $U(x, y, t)$ of clusters characterised by two “continuous” size coordinates x (number of X -type atoms) and y (number of Y -type atoms). In the large-cluster / continuum limit, U satisfies a drift–diffusion (Fokker–Planck) equation

$$\partial_t U = -\partial_x(F_x U) - \partial_y(F_y U) + \partial_{xx}(D_{xx} U) + \partial_{xy}(D_{xy} U) + \partial_{yy}(D_{yy} U). \quad (1)$$

Drift (F_x, F_y) and diffusion (D_{xx}, D_{yy}, D_{xy}) are derived from the master equation for a fixed set $\mathcal{M}$ of “mobile” species. Each $\mu = (m, q) \in \mathcal{M}$ adds or removes (m, q) atoms at a rate controlled by an absorption rate $\beta_\mu C_\mu(t)$ and an emission rate α_μ . Setting

$$d_\mu := \beta_\mu C_\mu + \alpha_\mu, \quad \gamma_\mu := \beta_\mu C_\mu - \alpha_\mu, \quad (2)$$

the standard truncation gives

$$F_x = \sum_{\mu \in \mathcal{M}} m \gamma_\mu, \quad F_y = \sum_{\mu \in \mathcal{M}} q \gamma_\mu, \quad (3)$$

$$D_{xx} = \frac{1}{2} \sum_{\mu \in \mathcal{M}} m^2 d_\mu, \quad D_{yy} = \frac{1}{2} \sum_{\mu \in \mathcal{M}} q^2 d_\mu, \quad D_{xy} = \sum_{\mu \in \mathcal{M}} m q d_\mu. \quad (4)$$

Here the off-diagonal D_{xy} is twice the matrix off-diagonal of $\mathbf{D} = \begin{pmatrix} D_{xx} & D_{xy}/2 \\ D_{xy}/2 & D_{yy} \end{pmatrix}$.

1.2 Why $d_\mu \geq 0$ on physical grounds

The non-negativity of d_μ is the foundation on which all subsequent stability and positivity arguments rest. It can be seen when studying the formulas for α and β or by physical argument. I thought it would be valuable to explain it in detail (to include this later in our report):

- $\beta_\mu \geq 0$ (**absorption rate constant**). β_μ multiplied by C_μ gives the per-cluster rate at which a free μ is captured by a cluster. A rate is the expected number of events per unit time, an intrinsically non-negative count. Else, it would be an emission, which is already captured by our emission term.
- $\alpha_\mu \geq 0$ (**emission rate constant**). Same argument: α_μ is the rate of emission of a μ -mobile from a cluster. Rates are non-negative.
- $C_\mu \geq 0$ (**concentration**). C_μ is a number density of free μ -clusters. Negative number densities are unphysical.

Combining: $d_\mu = \beta_\mu C_\mu + \alpha_\mu$ is a sum of two non-negative quantities, hence $d_\mu \geq 0$.

1.3 Positive semi-definiteness of the diffusion tensor

The PSD property of $\mathbf{D}$ follows immediately from $d_\mu \geq 0$. For any $\mathbf{v} = (v_x, v_y)^\top \in \mathbb{R}^2$:

$$\begin{aligned} \mathbf{v}^\top \mathbf{D} \mathbf{v} &= D_{xx} v_x^2 + D_{xy} v_x v_y + D_{yy} v_y^2 \\ &= \frac{1}{2} \sum_{\mu} d_\mu (m^2 v_x^2 + 2mq v_x v_y + q^2 v_y^2) = \frac{1}{2} \sum_{\mu} d_\mu (m v_x + q v_y)^2 \geq 0. \end{aligned} \quad (5)$$

The last expression is a sum of squares with non-negative weights d_μ . Equivalently:

$$D_{xx} \geq 0, \quad D_{yy} \geq 0, \quad D_{xy}^2 \leq 4 D_{xx} D_{yy}. \quad (6)$$

Where we used that a PSD matrix has a positive determinant for the third inequality.

Why this matters for the scheme. The PSD property is a physical fact about the master equation, not a numerical choice. No matter how exotic the mobile set $\mathcal{M}$ becomes, this chain of reasoning is unaffected.

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2 The diagonal stencil and mesh-adapted splitting (uniform grid, specific kind of model)

Throughout this section the grid is uniform with spacings Δx , Δy and aspect ratio $r := \Delta x / \Delta y > 0$. Cell centres are (x_i, y_j) .

2.1 The NE–SW diagonal stencil

Take the corner points $U_{i+1,j+1}$ and $U_{i-1,j-1}$ and Taylor-expand:

$$U_{i+1,j+1} = U_{ij} + \Delta x U_x + \Delta y U_y + \frac{\Delta x^2}{2} U_{xx} + \Delta x \Delta y U_{xy} + \frac{\Delta y^2}{2} U_{yy} + \mathcal{O}(h^3), \quad (7)$$

$$U_{i-1,j-1} = U_{ij} - \Delta x U_x - \Delta y U_y + \frac{\Delta x^2}{2} U_{xx} + \Delta x \Delta y U_{xy} + \frac{\Delta y^2}{2} U_{yy} + \mathcal{O}(h^3). \quad (8)$$

The cross term $\Delta x \Delta y U_{xy}$ has the same sign in both expansions: Δx and Δy flip together, so their product does not. Adding and dividing by $\Delta x \Delta y$:

$$\mathcal{T}_h(U) := \frac{U_{i+1,j+1} - 2U_{ij} + U_{i-1,j-1}}{\Delta x \Delta y} = r \partial_{xx} U + 2 \partial_{xy} U + r^{-1} \partial_{yy} U + \mathcal{O}(h^2). \quad (9)$$

2.2 Mesh-adapted splitting

Look for non-negative coefficients $\tilde{A}_x, \tilde{A}_y, \tilde{D}_2$ such that

$$\text{Diff}(C) = \partial_{xx}(\tilde{A}_x C) + \partial_{yy}(\tilde{A}_y C) + \tilde{D}_2 \mathcal{T}_h(C). \quad (10)$$

Substituting (9) and matching coefficients of each second derivative:

$$\tilde{D}_2 = \frac{D_{xy}}{2}, \quad \tilde{A}_x = D_{xx} - r \tilde{D}_2, \quad \tilde{A}_y = D_{yy} - \frac{\tilde{D}_2}{r}. \quad (11)$$

2.3 Why we want all three coefficients non-negative (M-matrix property)

Every piece of the splitting (10) produces a coupling from the centre cell (i, j) to its neighbours with sign equal to the sign of its coefficient. Concretely, when we write the discrete diffusion operator $L_h^{\mathcal{D}}$ as a sparse matrix, the off-diagonal entries are:

- $+\tilde{A}_x / \Delta x^2$ at the axial neighbours $(i \pm 1, j)$,
- $+\tilde{A}_y / \Delta y^2$ at the axial neighbours $(i, j \pm 1)$,
- $+\tilde{D}_2 / (\Delta x \Delta y)$ at the diagonal corners $(i \pm 1, j \pm 1)$.

For our IMEX time stepping we have to invert $A := I - \Delta t L_h^{\mathcal{D}}$ at each step. The off-diagonal entries of A are then $-\Delta t$ times the entries above, so they are ≤ 0 exactly when $\tilde{A}_x, \tilde{A}_y, \tilde{D}_2 \geq 0$.

A matrix with positive diagonal, non-positive off-diagonals, and weak diagonal dominance is called an **M-matrix**. The property we need is that an M-matrix has $A^{-1} \geq 0$ entry-wise. This means: if $U^n \geq 0$ everywhere, then $U^{n+1} = A^{-1}(U^n + \dots) \geq 0$ everywhere too. So the M-matrix property is exactly what guarantees that the discrete update preserves non-negativity of U . If even one of $\tilde{A}_x, \tilde{A}_y, \tilde{D}_2$ is negative, A loses the M-matrix property and U^{n+1} can become negative, which is unphysical.

3 General non-negativity criterion

Last week we proposed bounds on r , limiting on how we can define our uniform meshes. **In the call we got the proposition to generalize this to any model $\mathcal{M}$.**

Claim. For arbitrary (D_{xx}, D_{yy}, D_{xy}) satisfying the PSD property (6), the splitting (11) has all three coefficients non-negative if and only if

$$\boxed{\frac{|D_{xy}|}{2 D_{yy}} \leq r \leq \frac{2 D_{xx}}{|D_{xy}|}}, \quad (12)$$

and this interval is non-empty for every physically admissible tensor.

We must carefully handle the two possible signs of D_{xy} , as we have seen in (6) that it does not have to be positive. This is something new compared to the last weeks.

3.1 Bounds on r when $D_{xy} \geq 0$

Substituting $\tilde{D}_2 = D_{xy}/2 \geq 0$ into (11):

- $\tilde{A}_x \geq 0 \iff D_{xx} - r D_{xy}/2 \geq 0 \iff r \leq 2 D_{xx}/D_{xy}.$
- $\tilde{A}_y \geq 0 \iff D_{yy} - D_{xy}/(2r) \geq 0 \iff r \geq D_{xy}/(2 D_{yy}).$

Combining,

$$\frac{D_{xy}}{2 D_{yy}} \leq r \leq \frac{2 D_{xx}}{D_{xy}}, \quad (13)$$

and this interval is non-empty iff $D_{xy}^2 \leq 4 D_{xx} D_{yy}$, which is exactly (6), which holds unconditionally by Section 1.3. The interval is therefore non-empty for every $\mathcal{M}$.

3.2 Handling $D_{xy} < 0$: anti-diagonal stencil

If $D_{xy} < 0$ (e.g. a species $(1, -1) \in \mathcal{M}$ with $mq < 0$), the NE–SW stencil gives $\tilde{D}_2 < 0$, which breaks the M-matrix property of Section 2.3. Use the NW–SE diagonal instead. The Taylor expansions of $U_{i-1,j+1}$ and $U_{i+1,j-1}$ have $\Delta x \Delta y U_{xy}$ terms of opposite sign:

$$\mathcal{T}_h^{\text{AD}}(U) := \frac{U_{i-1,j+1} - 2 U_{ij} + U_{i+1,j-1}}{\Delta x \Delta y} = r \partial_{xx} U - 2 \partial_{xy} U + r^{-1} \partial_{yy} U + \mathcal{O}(h^2). \quad (14)$$

Repeating the matching procedure with $\mathcal{T}_h^{\text{AD}}$ gives $\tilde{D}_2 = |D_{xy}|/2 \geq 0$ and the same axial formulas; the unified bound (12) covers both cases. (This new diagonal direction is not currently implemented in the code, but would be a very simple addition, making the scheme very general.)

3.3 Verification on two concrete models

Old model $\mathcal{M} = \{(0, 1), (1, 1)\}$ with equal rates $d_{(0,1)} = d_{(1,1)} = d$. $D_{xx} = d/2$, $D_{yy} = d$, $D_{xy} = d$, giving $\frac{1}{2} \leq r \leq 1$.

Current model $\mathcal{M} = \{(1, 0), (1, 1)\}$. Writing $d_1 \equiv d_{(1,0)}$, $d_2 \equiv d_{(1,1)}$:

$$D_{xx} = \frac{d_1 + d_2}{2}, \quad D_{yy} = \frac{d_2}{2}, \quad D_{xy} = d_2, \quad (15)$$

$$1 \leq r \leq 1 + \frac{d_1}{d_2}. \quad (16)$$

The formulas thus agree with what we did in the last weeks.

4 Comparison with the standard 9-point stencil

During our experiments, I noticed that it is very hard to create test-cases that make our initial 9-point stencil fail. I mostly only saw that we sometimes got negative concentrations. In this chapter, I lay out why the 9-point scheme is not positivity-preserving but is indeed numerically stable.

4.1 The 9-point stencil

From the very beginnings of the project we know that the standard centered discretization of the cross derivative is

$$\partial_{xy}^{(9)}U := \frac{U_{i+1,j+1} - U_{i+1,j-1} - U_{i-1,j+1} + U_{i-1,j-1}}{4 \Delta x \Delta y}, \quad (17)$$

and the full discrete operator is

$$\begin{aligned} (L_h^{(9)}U)_{ij} = & D_{xx} \frac{U_{i+1,j} - 2U_{ij} + U_{i-1,j}}{\Delta x^2} + D_{yy} \frac{U_{i,j+1} - 2U_{ij} + U_{i,j-1}}{\Delta y^2} \\ & + D_{xy} \partial_{xy}^{(9)}U. \end{aligned} \quad (18)$$

4.2 Stencil patterns and Fourier symbols

The mathematical heart of the comparison is best seen by writing both stencils as 3×3 weight patterns and computing their Fourier symbols directly. This subsection follows the ideas of Mr. Monasse's notes (from a few calls ago). In Section 5 I also outlined a full Fourier analysis (with more bookkeeping) for this, but this chapter just shows it in the easy way we discussed last time already.

We work on a cubic grid $\Delta x = \Delta y = h$ for clarity; the asymmetric case should work very similarly because we perform the operations focusing on single cells and their surroundings.

Stencil pattern of the 9-point operator

$$h^2 L_h^{(9)} \sim \begin{pmatrix} -D_{xy}/4 & +D_{yy} & +D_{xy}/4 \\ +D_{xx} & -2(D_{xx} + D_{yy}) & +D_{xx} \\ +D_{xy}/4 & +D_{yy} & -D_{xy}/4 \end{pmatrix}. \quad (19)$$

For $D_{xy} > 0$, the NW and SE corners are **negative**; for $D_{xy} < 0$, the NE and SW corners are negative. One pair of opposite corners always has the wrong sign. This is the M-matrix violation visible directly in the stencil.

Stencil pattern of the diagonal-split operator

For the NE–SW branch ($\sigma = +1$, $D_{xy} \geq 0$), $L_h^{\text{split}} = \tilde{A}_x \partial_{xx}^{(5)} + \tilde{A}_y \partial_{yy}^{(5)} + \tilde{D}_2 \mathcal{T}_h$ gives

$$h^2 L_h^{\text{split}} \sim \begin{pmatrix} 0 & +\tilde{A}_y & +\tilde{D}_2 \\ +\tilde{A}_x & -2(\tilde{A}_x + \tilde{A}_y + \tilde{D}_2) & +\tilde{A}_x \\ +\tilde{D}_2 & +\tilde{A}_y & 0 \end{pmatrix}. \quad (20)$$

NW and SE corners are zero; the cross information lives only on the NE–SW corners with non-negative coefficient $\tilde{D}_2 = D_{xy}/2$. Under the unified bound (12), all three of $\tilde{A}_x, \tilde{A}_y, \tilde{D}_2$ are non-negative, so **every off-diagonal entry is ≥ 0** .

Fourier symbol of the full 9-point operator

We compute the symbol of the entire $L_h^{(9)}$ operator at once (like done with Monasse in call). Plug $C_{j,k} = \alpha e^{i(j\xi+k\eta)}$ into the full operator times h^2 :

$$h^2(L_h^{(9)}C)_{jk} = D_{xx}(C_{j+1,k} - 2C_{j,k} + C_{j-1,k}) + D_{yy}(C_{j,k+1} - 2C_{j,k} + C_{j,k-1}) \\ + \frac{D_{xy}}{4}(C_{j+1,k+1} - C_{j+1,k-1} - C_{j-1,k+1} + C_{j-1,k-1}).$$

Each shift in j is a multiplicative factor $e^{\pm i\xi}$, each shift in k a factor $e^{\pm i\eta}$. Dividing through by $C_{j,k}$ leaves the symbol:

$$h^2\widehat{L_h^{(9)}}(\xi, \eta) = D_{xx}(e^{i\xi} - 2 + e^{-i\xi}) + D_{yy}(e^{i\eta} - 2 + e^{-i\eta}) \\ + \frac{D_{xy}}{4}(e^{i(\xi+\eta)} - e^{i(\xi-\eta)} - e^{-i(\xi-\eta)} + e^{-i(\xi+\eta)}).$$

Pair complex conjugates using $e^{i\theta} + e^{-i\theta} = 2\cos\theta$:

$$h^2\widehat{L_h^{(9)}} = D_{xx}(2\cos\xi - 2) + D_{yy}(2\cos\eta - 2) + \frac{D_{xy}}{4}(2\cos(\xi + \eta) - 2\cos(\xi - \eta)).$$

Now apply the two trigonometric identities:

- Half-angle: $2\cos\theta - 2 = -4\sin^2(\theta/2)$ [from $1 - \cos\theta = 2\sin^2(\theta/2)$].
- Sum-to-product: $\cos(A + B) - \cos(A - B) = -2\sin A \sin B$.

Applying these to the three pieces:

$$D_{xx}(2\cos\xi - 2) = -4D_{xx}\sin^2(\xi/2), \\ D_{yy}(2\cos\eta - 2) = -4D_{yy}\sin^2(\eta/2), \\ \frac{D_{xy}}{4}(2\cos(\xi + \eta) - 2\cos(\xi - \eta)) = \frac{D_{xy}}{2} \cdot (-2\sin\xi \sin\eta) = -D_{xy}\sin\xi \sin\eta.$$

Putting it together:

$$\widehat{L_h^{(9)}}(\xi, \eta) = -\frac{4D_{xx}}{h^2}\sin^2\frac{\xi}{2} - \frac{4D_{yy}}{h^2}\sin^2\frac{\eta}{2} - \frac{D_{xy}}{h^2}\sin\xi \sin\eta. \quad (21)$$

The third term has **indefinite sign**: $\sin\xi \sin\eta > 0$ on modes oscillating along the NE–SW diagonal (ξ, η same sign), < 0 on modes along the NW–SE diagonal (ξ, η opposite signs). This is the “no sign due to sine–sine” issue presented by Mr. Monasse.

Fourier symbol of the full diagonal-split operator

We do the same procedure for $L_h^{\text{split}} = \tilde{A}_x\partial_{xx}^{(5)} + \tilde{A}_y\partial_{yy}^{(5)} + \tilde{D}_2\mathcal{T}_h$. Apply $h^2L_h^{\text{split}}$ to $C_{j,k} = \alpha e^{i(j\xi+k\eta)}$:

$$h^2(L_h^{\text{split}}C)_{jk} = \tilde{A}_x(C_{j+1,k} - 2C_{j,k} + C_{j-1,k}) + \tilde{A}_y(C_{j,k+1} - 2C_{j,k} + C_{j,k-1}) \\ + \tilde{D}_2(C_{j+1,k+1} - 2C_{j,k} + C_{j-1,k-1}).$$

After dividing through by $C_{j,k}$:

$$h^2\widehat{L_h^{\text{split}}}(\xi, \eta) = \tilde{A}_x(e^{i\xi} - 2 + e^{-i\xi}) + \tilde{A}_y(e^{i\eta} - 2 + e^{-i\eta}) \\ + \tilde{D}_2(e^{i(\xi+\eta)} - 2 + e^{-i(\xi+\eta)}).$$

Each parenthesis has the same shape $e^{i\theta} - 2 + e^{-i\theta}$, with $\theta = \xi, \eta, \xi + \eta$ respectively. By the basic identity

$$e^{i\theta} - 2 + e^{-i\theta} = 2 \cos \theta - 2 = -4 \sin^2(\theta/2),$$

each piece becomes a single $\sin^2$ term:

$$\widehat{L_h^{\text{split}}}(\xi, \eta) = -\frac{4\tilde{A}_x}{h^2} \sin^2 \frac{\xi}{2} - \frac{4\tilde{A}_y}{h^2} \sin^2 \frac{\eta}{2} - \frac{4\tilde{D}_2}{h^2} \sin^2 \frac{\xi + \eta}{2}. \quad (22)$$

This is a sum of three non-positive $\sin^2$ terms, each multiplied by a non-negative coefficient (under the unified bound (12)). The whole symbol is therefore ≤ 0 , so no sign issue.

What this means for stability

A scheme is called **L²-stable** if no Fourier mode is amplified from one time step to the next. To see what this requires from the symbol $\widehat{L}_h$, consider a single forward Euler step applied to one Fourier mode: $U_{ij}^{n+1} = U_{ij}^n + \Delta t (L_h U^n)_{ij}$. Since L_h acts on the mode by multiplication with $\widehat{L}_h$, the amplification factor is

$$G = 1 + \Delta t \widehat{L}_h.$$

For stability we need $|G| \leq 1$. When $\widehat{L}_h$ is real (which it is for our diffusion operators, because all three terms in (21) and (22) are real), G is just a real number, and $|G| \leq 1$ requires $-2 \leq \Delta t \widehat{L}_h \leq 0$. The right inequality $\Delta t \widehat{L}_h \leq 0$ holds for any $\Delta t > 0$ exactly when $\widehat{L}_h(\xi, \eta) \leq 0$ for every (ξ, η) . Further implications on CFL condition are discussed in Section 5.

- **Both schemes are L²-stable under PSD.** Under $D_{xy}^2 \leq 4D_{xx}D_{yy}$, the overall sign of $\widehat{L}_h^{(9)}$ is still ≤ 0 on the entire (ξ, η) box: the favourable axial $\sin^2$ pieces dominate the unfavourable $\sin^2((\xi - \eta)/2)$ piece. This was the reason why we can not easily create exploding examples under our physical assumptions $D_{xy}^2 \leq 4D_{xx}D_{yy}$. The explicit proof is given in the next subsection.
- **Only the diagonal-split scheme is positivity-preserving.** The 9-point's wrong-sign contribution along the NW–SE direction shows up as wrong-sign matrix entries at the NW and SE corners of $L_h^{(9)}$. After multiplying by a non-negative function and applying one explicit time step, those matrix entries produce negative pointwise values at the NW and SE corner offsets, clearly violating the discrete maximum principle.
- **When it is still unstable:** As the L²-stability is given under PSD (which internally requires $d_\mu \geq 0$), we could mathematically propably produce an instability by exploiting the possible non-positivities in the 9-point stencil. As shown in Section 7 this is basically impossible for our naive implementation in an IMEX setup, but realistic in a fully explicit environment.

4.3 The 9-point scheme is L²-stable but not positivity-preserving

We can show the 9-point symbol (21) is ≤ 0 for every $(\xi, \eta) \in [-\pi, \pi]^2$ explicitly. Use the double-angle formula $\sin \theta = 2 \sin(\theta/2) \cos(\theta/2)$ to rewrite the cross term:

$$\sin \xi \sin \eta = 4 \sin(\xi/2) \cos(\xi/2) \sin(\eta/2) \cos(\eta/2).$$

Multiplying (21) by $-h^2$:

$$-h^2 \widehat{L}_h^{(9)} = 4D_{xx} \sin^2(\xi/2) + 4D_{yy} \sin^2(\eta/2) + 4D_{xy} \sin(\xi/2) \cos(\xi/2) \sin(\eta/2) \cos(\eta/2).$$

Writing $a := \sin(\xi/2)$ and $b := \sin(\eta/2)$, and using $|\cos(\xi/2) \cos(\eta/2)| \leq 1$:

$$|D_{xy} \sin(\xi/2) \cos(\xi/2) \sin(\eta/2) \cos(\eta/2)| \leq |D_{xy}| |a| |b|.$$

Hence the worst case for non-positivity is

$$-h^2 \widehat{L_h^{(9)}} \geq 4(D_{xx} a^2 - |D_{xy}| |a| |b| + D_{yy} b^2).$$

The bracket is a quadratic in $|a|$ (with $|b|$ frozen). Complete the square:

$$D_{xx} a^2 - |D_{xy}| |a| |b| + D_{yy} b^2 = D_{xx} \left(|a| - \frac{|D_{xy}| |b|}{2D_{xx}} \right)^2 + b^2 \left(D_{yy} - \frac{D_{xy}^2}{4D_{xx}} \right).$$

The first term is ≥ 0 trivially. The second is ≥ 0 exactly when $D_{xy}^2 \leq 4D_{xx}D_{yy}$, which is the PSD bound (6), which we showed it holds physically in Section 1.3. So $-h^2 \widehat{L_h^{(9)}} \geq 0$ everywhere, i.e. $\widehat{L_h^{(9)}}(\xi, \eta) \leq 0$ for every (ξ, η) . **The 9-point scheme is therefore L^2 -stable.**

The problem is the M-matrix property. The off-diagonal entries of $L_h^{(9)}$ at the four corners are

$$[L_h^{(9)}]_{ij, i\pm 1, j\pm 1} = \pm \frac{D_{xy}}{4 \Delta x \Delta y}. \quad (23)$$

For $D_{xy} > 0$ the NE-SW corners get $+D_{xy}/(4\Delta x \Delta y) > 0$ (good for M-matrix as discussed in Section 2.3) but the NW-SE corners get $-D_{xy}/(4\Delta x \Delta y) < 0$, which has the wrong sign. Once we form $A = I - \Delta t L_h^{(9)}$ for the implicit step, those NW and SE entries of A become positive, violating the off-diagonal-non-positive requirement of an M-matrix. Then A^{-1} can have negative entries, and $U^n \geq 0$ does not guarantee $U^{n+1} \geq 0$. The diagonal-split scheme avoids this by construction: zero NW/SE entries, non-negative NE/SW entries (or vice versa for $D_{xy} < 0$).

5 Von Neumann (Fourier) stability analysis, general case

Following standard numerical methods textbooks, Von Neumann analysis substitutes a single discrete Fourier mode $U_{ij}^n = G^n \exp(i(k_x i \Delta x + k_y j \Delta y))$ into the linearised update and asks whether the amplification factor $G \in \mathbb{C}$ satisfies $|G| \leq 1$ for every wavenumber. We work in the general case, arbitrary mobile set $\mathcal{M}$, D_{xy} of either sign, and use only the PSD property (6). We use a uniform mesh, but Section 6 shows that an extension to a non-uniform grid can be quite simple. I think most of the following assumptions should still hold then.

The end results will show the following:

1. The Fourier symbol of the split discrete diffusion operator is real and non-positive whenever $\tilde{A}_x, \tilde{A}_y, \tilde{D}_2 \geq 0$, or equivalently, whenever r lies in the unified bound (12).
2. Backward Euler is unconditionally stable for diffusion; forward Euler imposes $\Delta t \leq 2/\Lambda$ with the explicit constant $\Lambda = 4(\tilde{A}_x/\Delta x^2 + \tilde{A}_y/\Delta y^2 + \tilde{D}_2/(\Delta x \Delta y))$.
3. The IMEX scheme used in the code (implicit diffusion + explicit RK3 advection) retains only the advective CFL constraint.

5.1 Setup and ansatz

For the analysis we freeze the coefficients $F_x, F_y, D_{xx}, D_{yy}, D_{xy}$ at constants. The von Neumann result then applies locally; in the full simulation it is enforced cell by cell.

Define $\sigma := \text{sign}(D_{xy}) \in \{+1, -1\}$ (with $\sigma = +1$ if $D_{xy} = 0$ by convention) and write $\tilde{D}_2 := |D_{xy}|/2 \geq 0$. The diagonal-stencil index pair is $(\sigma_x, \sigma_y) = (\sigma, +1)$, covering both NE–SW ($\sigma = +1$) and NW–SE ($\sigma = -1$) branches. The discrete diffusion operator is

$$\begin{aligned} (L_h^D U)_{ij} = & \tilde{A}_x \frac{U_{i+1,j} - 2U_{ij} + U_{i-1,j}}{\Delta x^2} + \tilde{A}_y \frac{U_{i,j+1} - 2U_{ij} + U_{i,j-1}}{\Delta y^2} \\ & + \tilde{D}_2 \frac{U_{i+\sigma_x, j+\sigma_y} - 2U_{ij} + U_{i-\sigma_x, j-\sigma_y}}{\Delta x \Delta y}. \end{aligned} \quad (24)$$

Substitute the plane wave with phases $\xi := k_x \Delta x \in [-\pi, \pi]$, $\eta := k_y \Delta y \in [-\pi, \pi]$. A linear translation-invariant operator L_h acts as $(L_h U)_{ij} = \widehat{L}_h(\xi, \eta) U_{ij}$.

The basic identity.

$$e^{i\theta} + e^{-i\theta} - 2 = 2 \cos \theta - 2 = -4 \sin^2(\theta/2). \quad (25)$$

Two ingredients: $e^{\pm i\theta} = \cos \theta \pm i \sin \theta$ and $1 - \cos \theta = 2 \sin^2(\theta/2)$.

5.2 Symbols of building blocks

In Section 4.2 we already worked out the full operator symbols, here we do something similar, just block-by-block.

Axial second differences. Substitute the plane wave $U_{ij} = \widehat{U} G^n e^{i(\xi i + \eta j)}$ into the axial- x second difference. Using $U_{i\pm 1, j} = e^{\pm i\xi} U_{ij}$:

$$\frac{U_{i+1,j} - 2U_{ij} + U_{i-1,j}}{\Delta x^2} = \frac{e^{i\xi} - 2 + e^{-i\xi}}{\Delta x^2} U_{ij}.$$

Apply the basic identity (25) with $\theta = \xi$: $e^{i\xi} - 2 + e^{-i\xi} = -4 \sin^2(\xi/2)$. So:

$$\widehat{\partial_x^{(2)}}(\xi, \eta) = -\frac{4 \sin^2(\xi/2)}{\Delta x^2}, \quad \widehat{\partial_y^{(2)}}(\xi, \eta) = -\frac{4 \sin^2(\eta/2)}{\Delta y^2}. \quad (26)$$

Both are real and ≤ 0 for $\xi \in [-\pi, \pi]$, $\eta \in [-\pi, \pi]$.

Diagonal cross stencil (both branches). The diagonal stencil $\partial_d^{(2)}$ uses corners offset by $\pm(\sigma_x, \sigma_y)$. Applied to the plane wave:

$$U_{i\pm\sigma_x, j\pm\sigma_y} = e^{\pm i(\sigma_x\xi + \sigma_y\eta)} U_{ij}.$$

So

$$\frac{U_{i+\sigma_x, j+\sigma_y} - 2U_{ij} + U_{i-\sigma_x, j-\sigma_y}}{\Delta x \Delta y} = \frac{e^{i(\sigma_x\xi + \sigma_y\eta)} - 2 + e^{-i(\sigma_x\xi + \sigma_y\eta)}}{\Delta x \Delta y} U_{ij}.$$

Now use the basic identity (25) with $\theta = \sigma_x\xi + \sigma_y\eta$, giving $-4\sin^2((\sigma_x\xi + \sigma_y\eta)/2)/(\Delta x \Delta y)$.

In our two branches, $\sigma_y = +1$ in both, while $\sigma_x = \sigma$ (the sign of D_{xy}). So $\sigma_x\xi + \sigma_y\eta = \sigma\xi + \eta$. Since $\sin^2$ is even, $\sin^2((\sigma\xi + \eta)/2) = \sin^2((\xi + \sigma\eta)/2)$. The unified expression is:

$$\widehat{\partial_d^{(2)}}(\xi, \eta) = -\frac{4\sin^2((\xi + \sigma\eta)/2)}{\Delta x \Delta y}. \quad (27)$$

5.3 The combined diffusion symbol

Combining (26) and (27):

$$\widehat{L_h^{\mathcal{D}}}(\xi, \eta) = -\frac{4\tilde{A}_x}{\Delta x^2} \sin^2 \frac{\xi}{2} - \frac{4\tilde{A}_y}{\Delta y^2} \sin^2 \frac{\eta}{2} - \frac{4\tilde{D}_2}{\Delta x \Delta y} \sin^2 \frac{\xi + \sigma\eta}{2}. \quad (28)$$

A few observations follow immediately from this expression.

The symbol is real. Each of the three terms in (28) involves only $\sin^2$ of a real argument, multiplied by a real coefficient. So $\widehat{L_h^{\mathcal{D}}}(\xi, \eta) \in \mathbb{R}$ for every (ξ, η) and every PSD diffusion tensor.

Non-positivity from non-negative coefficients. If $\tilde{A}_x \geq 0$, $\tilde{A}_y \geq 0$, $\tilde{D}_2 \geq 0$, then each $\sin^2(\bullet) \geq 0$ multiplied by a non-negative coefficient gives a non-positive contribution. The sum of three non-positive numbers is non-positive, so $\widehat{L_h^{\mathcal{D}}}(\xi, \eta) \leq 0$ everywhere on $[-\pi, \pi]^2$.

This is a sufficient condition for $\widehat{L_h^{\mathcal{D}}} \leq 0$ which could in principle hold for some choices with one coefficient slightly negative.

Spectral bound. Under the non-negativity assumption,

$$0 \leq -\widehat{L_h^{\mathcal{D}}}(\xi, \eta) \leq \Lambda, \quad \Lambda := 4 \left(\frac{\tilde{A}_x}{\Delta x^2} + \frac{\tilde{A}_y}{\Delta y^2} + \frac{\tilde{D}_2}{\Delta x \Delta y} \right), \quad (29)$$

since each $\sin^2(\bullet) \leq 1$, so the magnitude of each piece is at most its coefficient.

Reformulation in tensor entries. Using (11), $\tilde{A}_x/\Delta x^2 + \tilde{D}_2/(\Delta x \Delta y) = D_{xx}/\Delta x^2$ (the cross-term cancels via $r\Delta y/\Delta x = 1$), and similarly for y . Hence

$$\Lambda = 4 \left(\frac{D_{xx}}{\Delta x^2} + \frac{D_{yy}}{\Delta y^2} - \frac{|D_{xy}|}{2\Delta x \Delta y} \right). \quad (30)$$

The cross-term enters with a negative sign: a strong $|D_{xy}|$ relaxes the spectral bound, because part of the diffusion is shared between axial and diagonal contributions.

5.4 Stability for time integrators

Stability means $|G| \leq 1$ for every wavenumber, also after using it on our time integrators.

Forward Euler. The forward Euler update is $U^{n+1} = U^n + \Delta t L_h U^n$. For a single Fourier mode every grid value picks up the same factor, so $G^{n+1} = G^n(1 + \Delta t \widehat{L}_h)$, giving

$$G_{\text{FE}} = 1 + \Delta t \widehat{L}_h.$$

For our diffusion operator the symbol $\widehat{L}_h^{\mathcal{D}}$ is real and ≤ 0 (we proved this above). So G_{FE} is a real number between $1 + \Delta t \widehat{L}_h^{\mathcal{D}}$ and 1. We need $|G_{\text{FE}}| \leq 1$, which means $-1 \leq 1 + \Delta t \widehat{L}_h^{\mathcal{D}}$, i.e. $\Delta t \cdot |\widehat{L}_h^{\mathcal{D}}| \leq 2$. The worst case (largest $|\widehat{L}_h^{\mathcal{D}}|$) is Λ from (29), so the condition becomes:

$$\Delta t \leq \frac{2}{\Lambda} = \frac{1}{2 \left(\frac{D_{xx}}{\Delta x^2} + \frac{D_{yy}}{\Delta y^2} - \frac{|D_{xy}|}{2 \Delta x \Delta y} \right)}. \quad (31)$$

Backward Euler. The backward Euler update is $U^{n+1} = U^n + \Delta t L_h U^{n+1}$, which means $(1 - \Delta t \widehat{L}_h) G^{n+1} = G^n$, giving

$$G_{\text{BE}} = \frac{1}{1 - \Delta t \widehat{L}_h}.$$

Since $\widehat{L}_h^{\mathcal{D}} \leq 0$, the denominator $1 - \Delta t \widehat{L}_h^{\mathcal{D}} \geq 1$, so $|G_{\text{BE}}| \leq 1$ for any $\Delta t > 0$: **unconditionally stable**. This is why we treat diffusion implicitly.

IMEX (advection explicit, diffusion implicit). The actual scheme in our code treats diffusion implicitly and advection explicitly in the same step:

$$(I - \Delta t L_h^{\mathcal{D}}) U^{n+1} = U^n + \Delta t L_h^{\mathcal{A}} U^n,$$

where $L_h^{\mathcal{A}}$ is the discrete advection operator. For a single Fourier mode, dividing both sides by G^n :

$$G_{\text{IMEX}} = \frac{1 + \Delta t \widehat{L}_h^{\mathcal{A}}}{1 - \Delta t \widehat{L}_h^{\mathcal{D}}}. \quad (32)$$

The denominator is real and ≥ 1 (same reasoning as backward Euler). So it can only shrink the magnitude: $|G_{\text{IMEX}}| \leq |1 + \Delta t \widehat{L}_h^{\mathcal{A}}|$. The diffusion contributes no constraint at all; only the advective numerator matters. With RK3 time stepping for the advection part, the stability region along the imaginary axis extends to $|\beta| \leq \sqrt{3}$, giving the CFL condition (reasoning follows below)

$$\Delta t \left(\frac{|F_x|}{\Delta x} + \frac{|F_y|}{\Delta y} \right) \leq \sqrt{3}. \quad (33)$$

To understand the CFL bound we need the advection symbol. For a centred first difference applied to the plane wave $U_{ij} = e^{i(\xi i + \eta j)}$:

$$\frac{U_{i+1,j} - U_{i-1,j}}{2\Delta x} = \frac{e^{i\xi} - e^{-i\xi}}{2\Delta x} U_{ij} = \frac{i \sin \xi}{\Delta x} U_{ij}, \quad (34)$$

where we used $e^{i\xi} - e^{-i\xi} = 2i \sin \xi$. The symbol is purely imaginary.

The full 2D advection symbol is therefore $\widehat{L}_h^{\mathcal{A}} = i(F_x \sin \xi / \Delta x + F_y \sin \eta / \Delta y) =: i\beta$ with $\beta \in \mathbb{R}$.

Now consider the third-order Runge–Kutta method (RK3). For a scalar ODE $u' = \lambda u$, one RK3 step gives $u^{n+1} = G(z) u^n$ with $z := \lambda \Delta t$ and the amplification polynomial

$$G(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6}.$$

In our case $\lambda = i\beta/\Delta t$ is purely imaginary, so $z = i\beta \Delta t \cdot (1/\Delta t) = i\beta$. Substituting $z = i\beta$ and using $(i\beta)^2 = -\beta^2$, $(i\beta)^3 = -i\beta^3$:

$$G(i\beta) = \underbrace{\left(1 - \frac{\beta^2}{2}\right)}_{\text{real part}} + i \underbrace{\left(\beta - \frac{\beta^3}{6}\right)}_{\text{imaginary part}}.$$

The squared modulus is

$$|G|^2 = \left(1 - \frac{\beta^2}{2}\right)^2 + \left(\beta - \frac{\beta^3}{6}\right)^2.$$

Expanding and collecting powers of β :

$$\begin{aligned} \left(1 - \frac{\beta^2}{2}\right)^2 &= 1 - \beta^2 + \frac{\beta^4}{4}, \\ \left(\beta - \frac{\beta^3}{6}\right)^2 &= \beta^2 - \frac{\beta^4}{3} + \frac{\beta^6}{36}. \end{aligned}$$

Adding:

$$|G|^2 = 1 + \beta^4\left(\frac{1}{4} - \frac{1}{3}\right) + \frac{\beta^6}{36} = 1 - \frac{\beta^4}{12} + \frac{\beta^6}{36}.$$

We need $|G|^2 \leq 1$, i.e. $\beta^4\left(-\frac{1}{12} + \frac{\beta^2}{36}\right) \leq 0$. Since $\beta^4 \geq 0$, this requires $\beta^2/36 \leq 1/12$, i.e. $\beta^2 \leq 3$, giving $|\beta| \leq \sqrt{3}$.

The largest value of $|\beta|$ over all wavenumbers is $|\beta|_{\max} = \Delta t(|F_x|/\Delta x + |F_y|/\Delta y)$ (since $|\sin \xi| \leq 1$ and $|\sin \eta| \leq 1$, and the worst case is when both hit 1 simultaneously). The stability condition $|\beta|_{\max} \leq \sqrt{3}$ therefore gives

$$\Delta t \left(\frac{|F_x|}{\Delta x} + \frac{|F_y|}{\Delta y} \right) \leq \sqrt{3}.$$

6 Non-uniform (geometric) grid

A geometric grid was proposed 2 weeks ago to be able to handle longer simulation-runs. A similar work to this chapter was conducted by other team members already in separate documents. To achieve a concrete implementation, further steps are needed; thus, this chapter revisits some of the calculations for reference.

6.1 Grid definition

Let:

$$\Delta x_{n+1} = (1 + \alpha) \Delta x_n, \quad \Delta y_{m+1} = (1 + \alpha) \Delta y_m, \quad \alpha \approx 0.03. \quad (35)$$

The same growth factor α in both directions is assumed for now. This choice is explained later and it seems to be necessary (**we could probably investigate this a bit more**).

Cell (i, j) occupies $\Omega_{ij} = [x_{i-1/2}, x_{i+1/2}] \times [y_{j-1/2}, y_{j+1/2}]$ with area $|\Omega_{ij}| = \Delta x_i \Delta y_j$. Half-distances between cell centres are

$$h_x^+ := x_{i+1} - x_i = \frac{1}{2}(\Delta x_i + \Delta x_{i+1}), \quad h_x^- := x_i - x_{i-1} = \frac{1}{2}(\Delta x_{i-1} + \Delta x_i), \quad (36)$$

and analogously $h_y^\pm$. On the geometric grid (35),

$$\frac{h_x^+}{h_x^-} = \frac{\Delta x_i + \Delta x_{i+1}}{\Delta x_{i-1} + \Delta x_i} = \frac{1 + (1 + \alpha)}{(1 + \alpha)^{-1} + 1} = 1 + \alpha, \quad (37)$$

and identically $h_y^+/h_y^- = 1 + \alpha$. Hence:

$$\boxed{\frac{h_x^+}{h_x^-} = \frac{h_y^+}{h_y^-} = 1 + \alpha.} \quad (38)$$

6.2 Axial diffusion fluxes

The $\partial_{xx}(\tilde{A}_x C)$ term in flux form (as shown last week):

$$G_{i+1/2,j}^x = \tilde{A}_{x,i+1/2,j} \frac{U_{i+1,j} - U_{i,j}}{h_x^+}, \quad \tilde{A}_{x,i+1/2,j} := \frac{1}{2}(\tilde{A}_{x,i,j} + \tilde{A}_{x,i+1,j}), \quad (39)$$

and the cell update

$$\text{diff}_{i,j}^{(x)} = \frac{G_{i+1/2,j}^x - G_{i-1/2,j}^x}{\Delta x_i}. \quad (40)$$

Each G^x is single-valued and enters two cells with opposite signs; mass conservation is automatic. The y -direction is identical.

6.3 Weighted diagonal stencil on the non-uniform grid

The naive three-point diagonal stencil is inconsistent on a non-uniform grid because the forward step h_x^+ differs from the backward step h_x^- . We need a weighted combination.

Taylor-expand:

$$U_{i+1,j+1} = U_{ij} + h_x^+ U_x + h_y^+ U_y + \frac{(h_x^+)^2}{2} U_{xx} + h_x^+ h_y^+ U_{xy} + \frac{(h_y^+)^2}{2} U_{yy} + \mathcal{O}(h^3), \quad (41)$$

$$U_{i-1,j-1} = U_{ij} - h_x^- U_x - h_y^- U_y + \frac{(h_x^-)^2}{2} U_{xx} + h_x^- h_y^- U_{xy} + \frac{(h_y^-)^2}{2} U_{yy} + \mathcal{O}(h^3). \quad (42)$$

For weights $w^+, w^- > 0$:

$$\begin{aligned} \Phi &:= w^+ U_{i+1,j+1} + w^- U_{i-1,j-1} - (w^+ + w^-) U_{ij} \\ &= (w^+ h_x^+ - w^- h_x^-) U_x + (w^+ h_y^+ - w^- h_y^-) U_y + (\text{2nd order}). \end{aligned} \quad (43)$$

The first-order terms vanish iff $w^+ h_x^+ = w^- h_x^-$ and $w^+ h_y^+ = w^- h_y^-$ simultaneously, requiring

$$\frac{w^-}{w^+} = \frac{h_x^+}{h_x^-} = \frac{h_y^+}{h_y^-}, \quad (44)$$

which is exactly (38).

Why both alpha need to be the same. If we picked $\alpha_x \neq \alpha_y$, then $h_x^+/h_x^- \neq h_y^+/h_y^-$ and no choice of weights $w^\pm$ can kill the first-order terms in (43) simultaneously. After the division by $\mathcal{N}_{ij}$ (see below), the second-derivative terms come out at $\mathcal{O}(1)$ because both numerator and denominator scale like h^3 . But the leftover first-order numerator scales like h^2 , so after dividing by $\mathcal{N}_{ij} \sim h^3$ the residual drift contribution scales as $\mathcal{O}(h^{-1})$ and diverges as the grid is refined. The discrete operator $\mathcal{T}_h^{\text{nu}}$ would no longer approximate the diffusion operator we want; it would approximate something with a spurious unbounded drift which we would have no terms to absorb.

Choosing $w^+ = h_x^-$, $w^- = h_x^+$ and using (38):

$$[\Phi]_{U_{xx}} = \frac{1}{2} h_x^+ h_x^- (h_x^+ + h_x^-), \quad (45)$$

$$[\Phi]_{U_{xy}} = h_x^+ h_x^- (h_y^+ + h_y^-), \quad (46)$$

$$[\Phi]_{U_{yy}} = \frac{1}{2} h_y^+ h_y^- (h_x^+ + h_x^-). \quad (47)$$

The normalisation $\mathcal{N}_{ij}$. We want the resulting non-uniform stencil to take the same form as the uniform-grid stencil (9): $\mathcal{T}_h^{\text{nu}}(U) = \tilde{r}_{ij} \partial_{xx} U + 2 \partial_{xy} U + \tilde{r}_{ij}^{-1} \partial_{yy} U + \mathcal{O}(h)$ because then the matching procedure (10) carries over verbatim with r replaced by the cell-local $\tilde{r}_{ij}$. The key requirement is that the coefficient of $\partial_{xy} U$ is exactly $+2$. Looking at the row above, the U_{xy} coefficient is $h_x^+ h_x^- (h_y^+ + h_y^-)$. To turn that into 2, we divide by $\mathcal{N}_{ij} := \frac{1}{2} h_x^+ h_x^- (h_y^+ + h_y^-)$. After this normalisation, the ∂_{xx} and ∂_{yy} coefficients automatically come out as $\tilde{r}_{ij}$ and $\tilde{r}_{ij}^{-1}$:

$$\boxed{\mathcal{T}_h^{\text{nu}}(U) = \tilde{r}_{ij} \partial_{xx} U + 2 \partial_{xy} U + \tilde{r}_{ij}^{-1} \partial_{yy} U + \mathcal{O}(h), \quad \tilde{r}_{ij} := \frac{h_x^+ + h_x^-}{h_y^+ + h_y^-}.} \quad (48)$$

This has exactly the same form as (9) with r replaced by the cell-local $\tilde{r}_{ij}$.

6.4 Splitting and local non-negativity

The splitting carries over verbatim with $r \rightarrow \tilde{r}_{ij}$:

$$\tilde{D}_{2,ij} = \frac{D_{xy,ij}}{2}, \quad \tilde{A}_{x,ij} = D_{xx,ij} - \tilde{r}_{ij} \tilde{D}_{2,ij}, \quad \tilde{A}_{y,ij} = D_{yy,ij} - \frac{\tilde{D}_{2,ij}}{\tilde{r}_{ij}}, \quad (49)$$

and the admissibility (12) becomes a cell-local condition

$$\frac{|D_{xy,ij}|}{2 D_{yy,ij}} \leq \tilde{r}_{ij} \leq \frac{2 D_{xx,ij}}{|D_{xy,ij}|} \quad \forall (i, j). \quad (50)$$

For $\mathcal{M} = \{(1, 0), (1, 1)\}$ this is our bound from last week $1 \leq \tilde{r}_{ij} \leq 1 + d_1[i]/d_2[i]$.

$\tilde{r}_{ij}$ on the geometric grid. Using $\Delta x_i = \Delta x_0(1 + \alpha)^i$, so

$$\tilde{r}_{ij} = \frac{\Delta x_i}{\Delta y_j} = \frac{\Delta x_0}{\Delta y_0} (1 + \alpha)^{i-j}. \quad (51)$$

Cells with large i at fixed j have larger $\tilde{r}$. The worst case for (50) is the corner cell with maximal $i - j$. This represents the cells in the grid which have the most irregular shapes.

7 Numerical Experiments: Breakdown of Positivity

Setup and goals

Both solvers use the same Strang splitting (half-advection $\rightarrow$ implicit diffusion $\rightarrow$ half-advection, MP5/SSP-RK3 advection at an advection-only CFL), making them L^2 -stable by construction if we choose physically feasible values. Their only structural difference is the matrix used in the implicit diffusion step: the **standard** solver assembles the 9-point stencil (D_{xx}, D_{yy}, D_{xy}) with negative NW/SE corner weights for $D_{xy} > 0$ (M-matrix violated), while the **diagonal** solver assembles the mesh-adapted split $(\tilde{A}_x, \tilde{A}_y, \tilde{D}_2)$ with exclusively non-negative off-diagonals (M-matrix satisfied). L^2 stability is shared, but positivity preservation is not.

Non-negativity is not only a physical requirement. The diffusion tensor itself is assembled from $d_1, d_2 > 0$ computed from positive concentrations: if the ODE cascade drives C_{10} or C_{11} below zero, the tensor loses positive semi-definiteness, making the linear system $(I - \Delta t L)$ potentially indefinite, a configuration under which even the IMEX implicit solve could fail. This theoretical route to instability in the diagonal solver **has not been reproduced numerically for our model parameters**, but it cannot be excluded a priori when positivity is violated upstream in the ODE. **We worked quite a lot on trying to find such an example, but could not come up with one in IMEX yet.**

The three cases keep both solvers implicit and run to $T = 10$ to validate that they agree. All cases use $r = 1$ ($\Delta x = \Delta y = 0.5$).

We have also conducted experiments using unphysical values for D_{xy} breaking our PSD assumption. In these cases the naive solver often shows numerical explosions, while the diagonal solver stays stable. We omitted these experiments, because they are not directly relevant to the problem we are solving.

Cases 1–3: implicit comparison (9-pt vs. diagonal, both Strang)

In these three cases both solvers use the identical Strang+implicit structure. The distributions produced are visually indistinguishable on the linear scale, confirming that the L^2 accuracy is equivalent and the implicit regularisation suppresses the M-matrix violation to machine-precision level. The only detectable difference is in the $\min(U)$ diagnostic: the 9-point implicit consistently produces a small strictly negative floor; the diagonal stays at floating-point positive throughout.

Case 1 – Extreme cross-coupling ($C_{11} = 5$, fixed mobile)

$\varepsilon = 1$, $C_{10} = 0.01$, $C_{11} = 5$ (fixed, `fixed_mobile=true`), $G_{10} = 0$; grid 177×69 ; Gaussian at (45, 18); $T = 10$. With $C_{11} = 5$ the off-diagonal $D_{xy} = d_2 \approx 5$ is twice the axial coefficients $D_{xx} \approx D_{yy} \approx 2.5$, creating the most adverse ratio for the M-matrix condition. Fixing the mobile concentrations keeps D_{xy} large throughout, preventing the natural equilibration that would otherwise reduce it. Both solvers complete the run; the snapshots are nearly identical.

Failure Case: extreme_coupling_spike

eps=1.00 C10=0.01 C11=5 G10=2 r=1.00 Nx=177 Ny=69

Standard=Strang implicit 9-pt (M-matrix violated) Diagonal=Strang implicit diagonal (M-matrix OK)

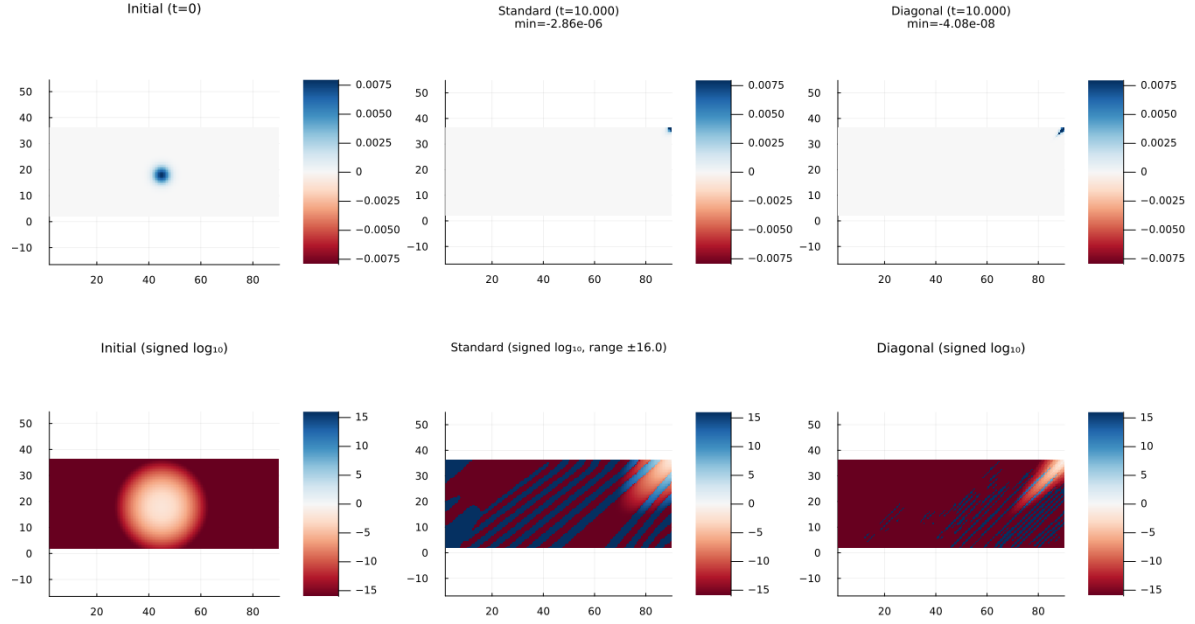


Figure 1: **Case 1 – Extreme cross-coupling (implicit comparison).** Top row: linear scale. Bottom row: signed $\log_{10}$ scale. Both methods reach $T = 10$ with indistinguishable distributions; the Gaussian has spread anisotropically along the NE–SW direction consistent with $D_{xy} > 0$. No visible negativity is present in either panel on the linear scale. The log-scale row shows the same smooth positive blob for both solvers, confirming that the implicit regularisation keeps the M-matrix violation below any observable threshold.

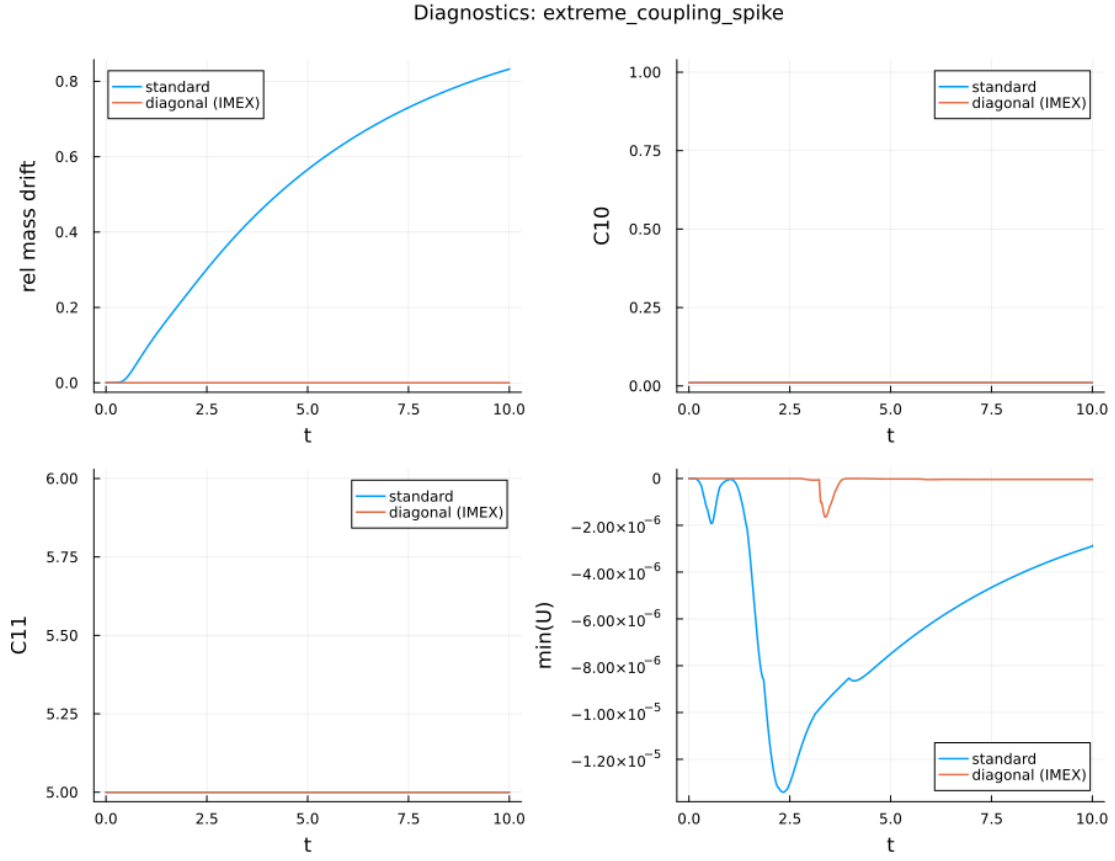


Figure 2: Diagnostics for Case 1. Mass drift, C_{10} , and C_{11} histories overlap between the two solvers, because the code forces them to stay the same in this example. The only discrepancy is in $\min(U)$: the 9-point implicit produces a persistent negative floor, while the diagonal stays at machine-precision positive.

Case 2 – Diagonal stripe ($G_{10} = 10^{-20}$)

$\varepsilon = 1$, $C_{10} = 0.05$, $C_{11} = 1.5$, $G_{10} = 10^{-20}$; grid 213×81 ; stripe along $x + y = 70$ (width $\sigma^2 = 1$); $T = 10$. The NW–SE stripe gradient aligns exactly with the direction of the negative 9-pt corner weights, representing the geometrically worst case for M-matrix violation. Despite this, both implicit solvers produce effectively the same result: the stripe spreads, decays under diffusion and advection, and the two distributions are again indistinguishable.

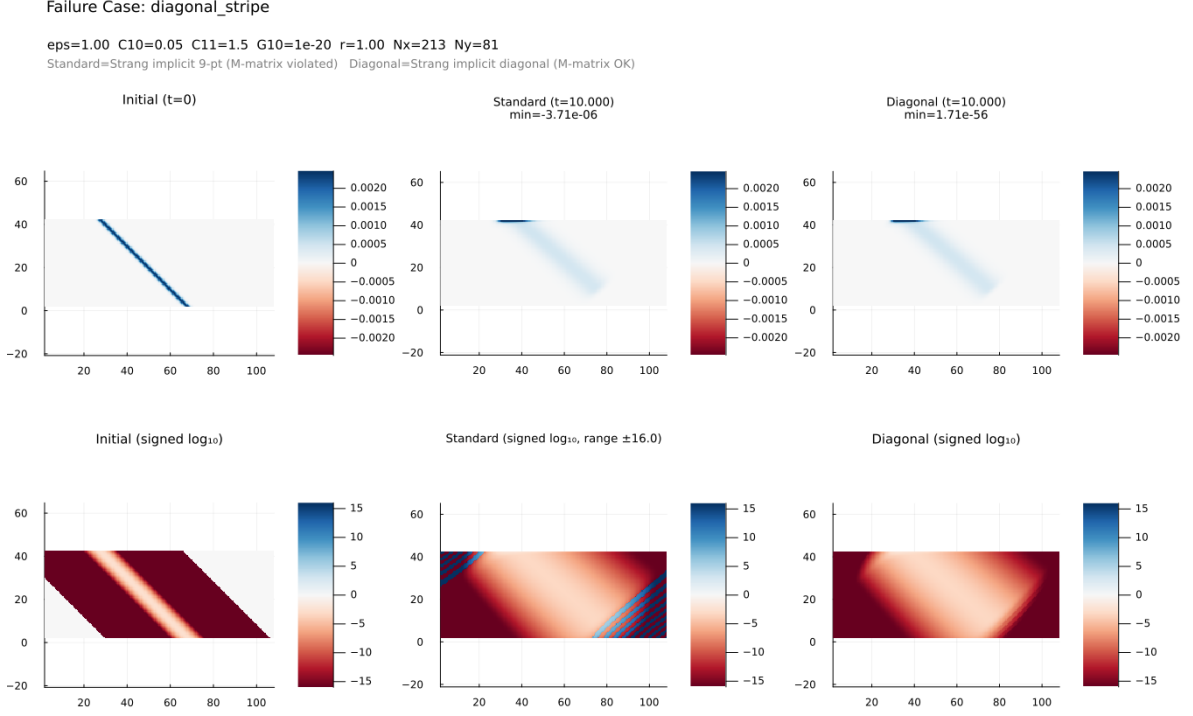


Figure 3: **Case 2 – Diagonal stripe (implicit comparison).** The initial NW–SE stripe is visible as a band at $t = 0$. By $T = 10$ the stripe has broadened symmetrically and its peak has drifted; both standard and diagonal panels show the same smooth distribution. The signed- $\log_{10}$ row reveals no meaningful difference, except the 9-point stencil producing some oscillatory patterns (at an extremely small scale).

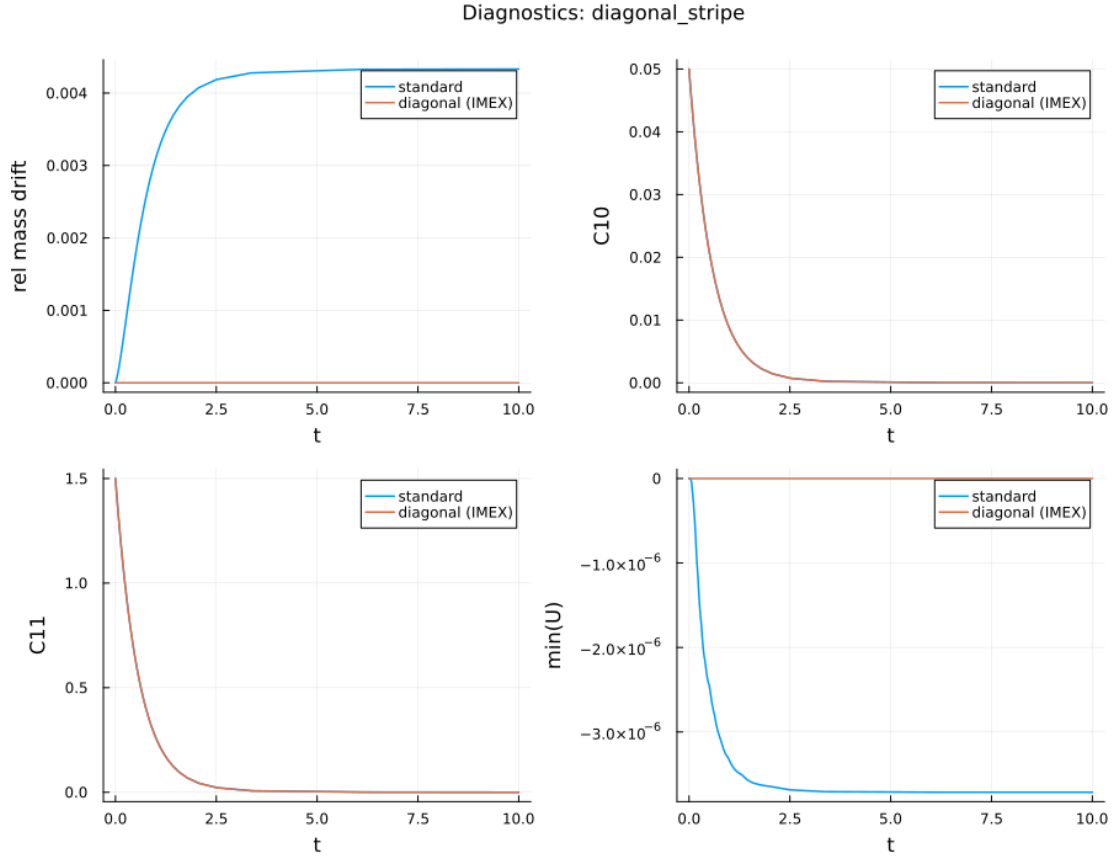


Figure 4: Diagnostics for Case 2. The solvers track each other closely throughout $T = 10$ in all four panels. C_{11} decays from 1.5 towards its equilibrium; $\min(U)$ in the 9-point solver shows small oscillating negatives ($\lesssim 10^{-20}$) that neither grow nor trigger the ODE cascade, confirming suppression by the implicit solve.

Case 3 – Single Gaussian blob ($G_{10} = 10^{-19}$, moderate coupling)

$\varepsilon = 1$, $C_{10} = 0.1$, $C_{11} = 1$, $G_{10} = 10^{-19}$; grid 177×69 ; Gaussian at $(35, 16)$; $T = 10$. Moderate parameters and a smooth initial condition represent a typical operating point. Both methods produce visually identical results throughout the run.

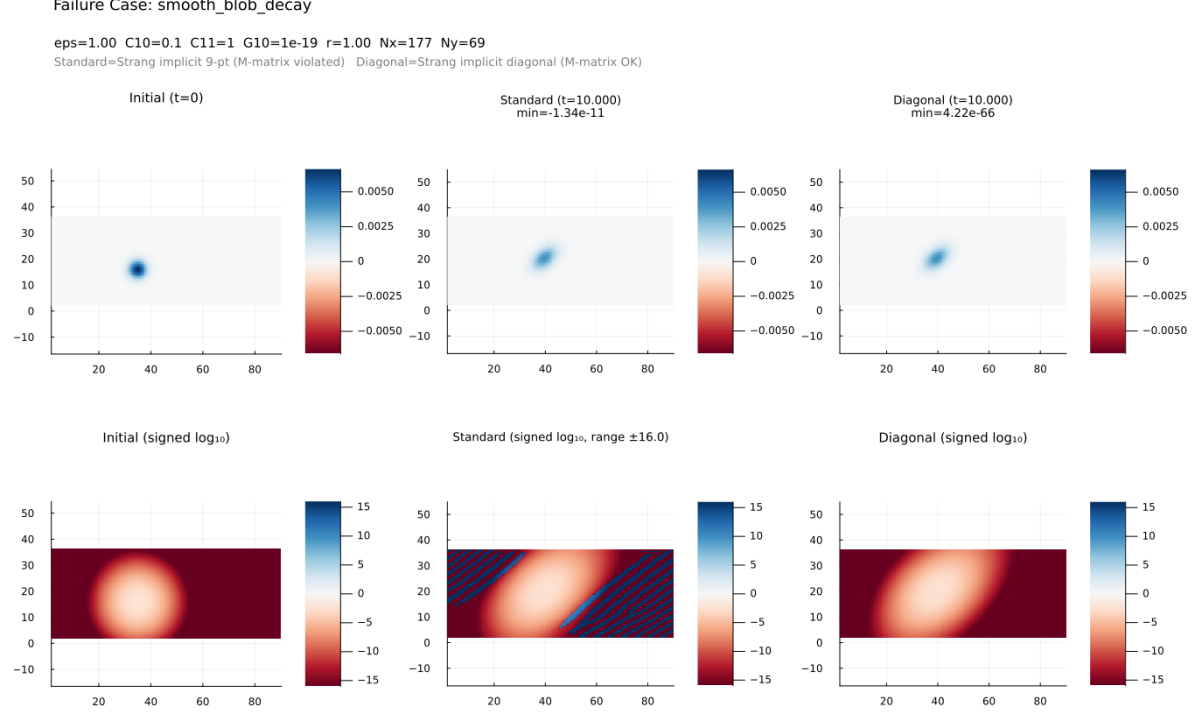


Figure 5: **Case 3 – Single blob (implicit comparison).** The blob spreads and its peak migrates under the combined drift and diffusion; standard and diagonal panels are indistinguishable on both linear and log scales (besides some minor oscillations). There is nothing to criticise in either solution at this parameter level.

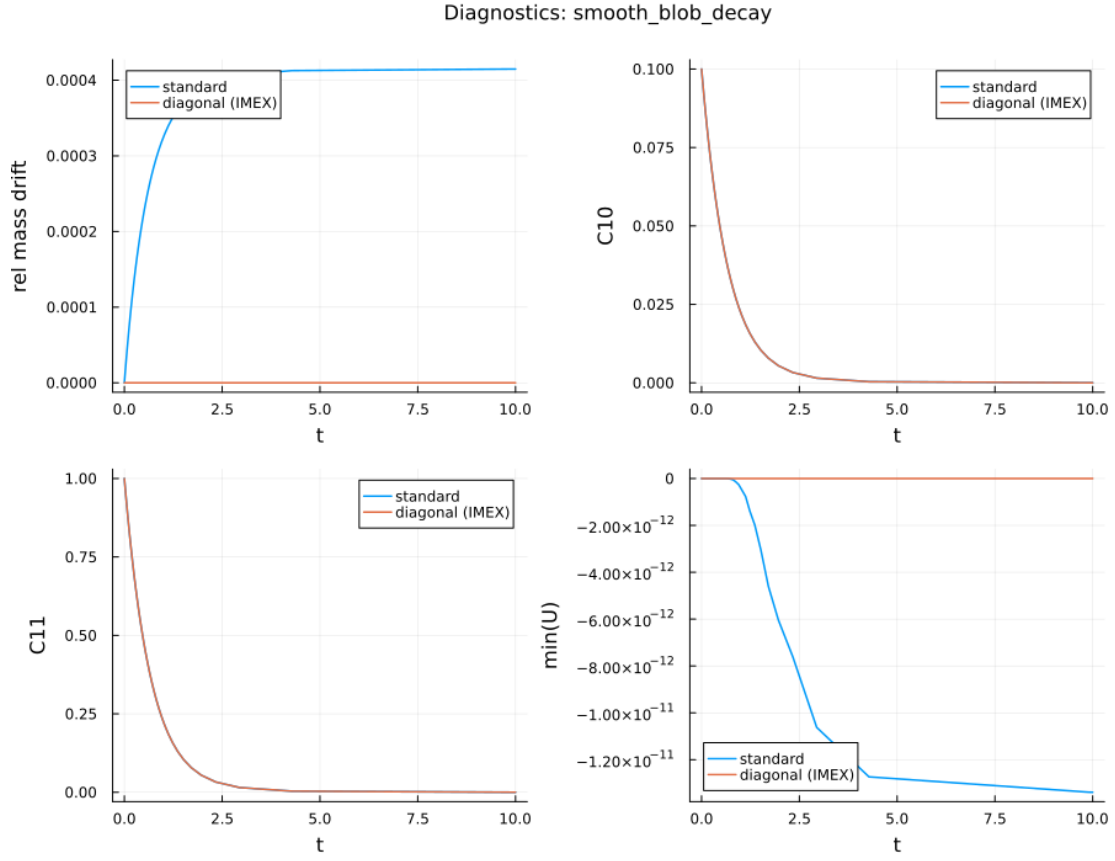


Figure 6: Diagnostics for Case 3. All four diagnostic traces coincide between the two solvers. $\min(U)$ in the 9-point solver remains at floating-point positive throughout; moderate $C_{11} = 1$ is insufficient to produce any detectable positivity violation in the implicit scheme.

8 Long-Horizon Experiments

All long-run experiments use the diagonal IMEX solver on a 501×201 grid ($\approx 100\,000$ unknowns, $\Delta x = \Delta y = 0.5$ unless noted) over $T = 1000$, with snapshots and diagnostics at every 10% checkpoint.

Experiment 1 – Slow G_{10} -driven irradiation growth

$\varepsilon = 1$, $C_{10}(0) = C_{11}(0) = 0.5$, $G_{10} = 10^{-18}$, $G_{11} = 0$, $r = 1$; Gaussian at $(10, 10)$, domain $[2, 252] \times [2, 102]$. The source $G_{10} = 10^{-18}$ is within the physical irradiation range. Both mobile concentrations equilibrate rapidly from their initial values of 0.5 to a tiny quasi-steady value. After this fast transient the drift velocity $F_x \propto C_{10}$ is negligible and the CFL allows $\Delta t \rightarrow \infty$; each remaining checkpoint is reached in a single step. The ten snapshots are therefore effectively identical after $t = 100$: the distribution barely moves, reflecting the physically correct quasi-static state. The experiment confirms mass conservation (rel. drift $< 10^{-10}$) over the full horizon without any visible numerical artefact.

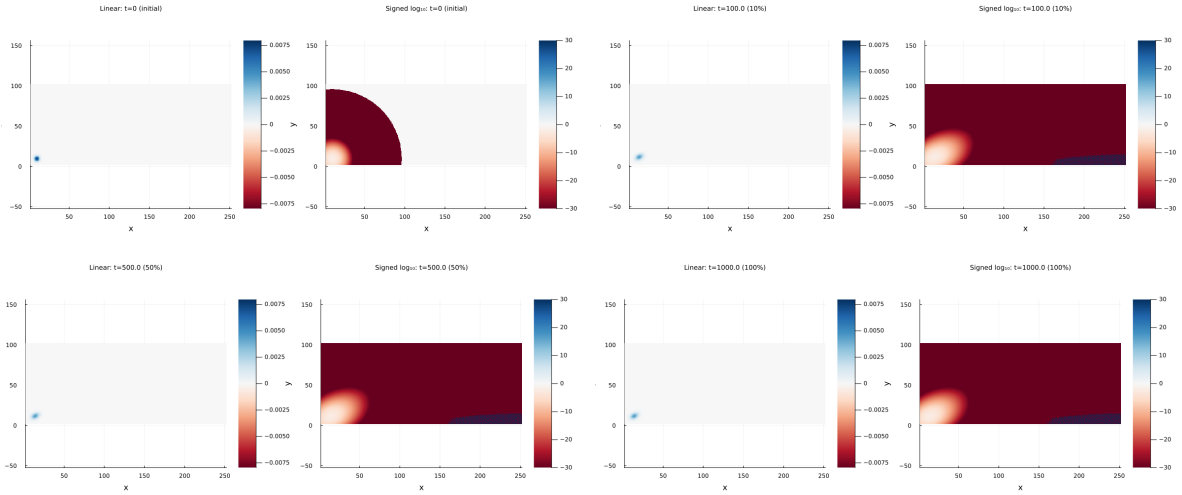


Figure 7: **Experiment 1 – Steady irradiation** ($G_{10} = 10^{-18}$). Snapshots at $t = 0, 100, 500, 1000$. After the first checkpoint the distribution is visually frozen: $C_{10}(1000) \approx 2.4 \times 10^{-6}$ is too small to drive any perceptible drift. The slight broadening compared to $t = 0$ reflects the diffusion during the initial equilibration phase only.

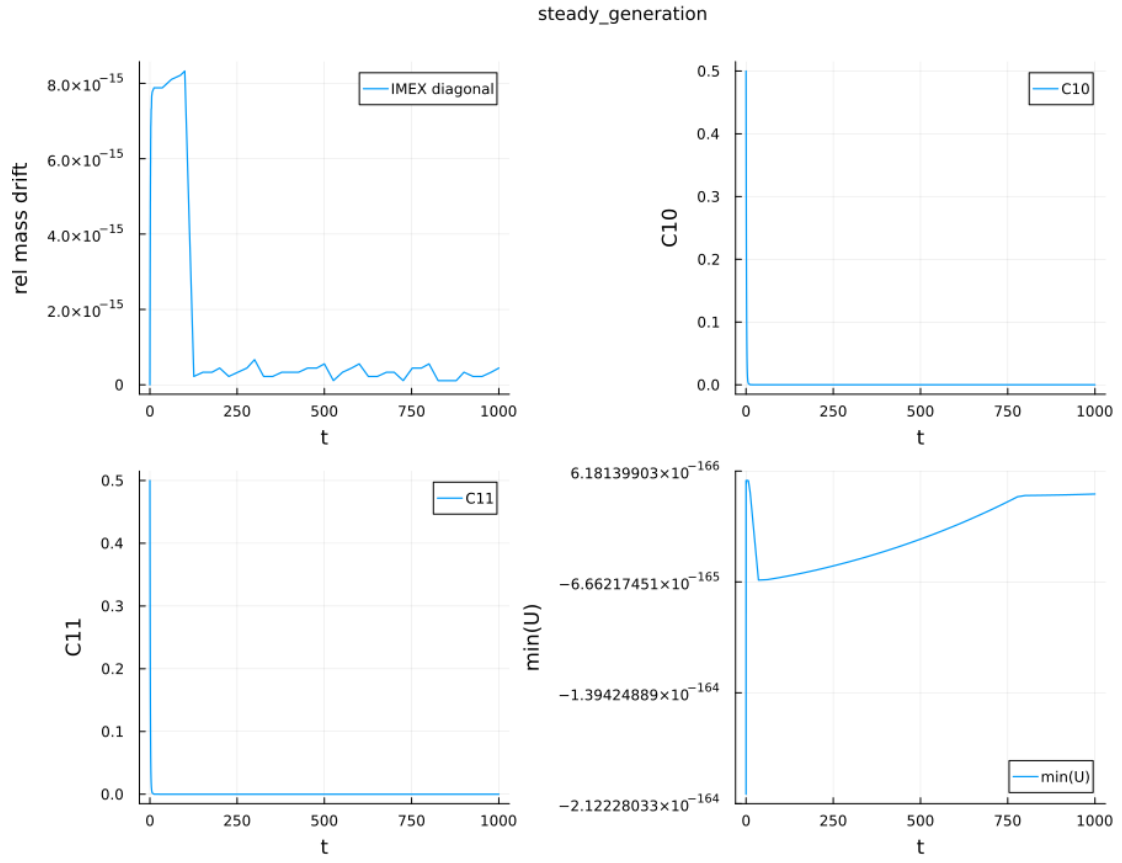


Figure 8: Diagnostics for Experiment 1.

Experiment 2 – Relaxation from large initial C_{10}

$\varepsilon = 0.5$, $C_{10}(0) = 5$, $C_{11}(0) = 0$, $G_{10} = G_{11} = 0$, $r = 1$; Gaussian at $(20, 20)$, domain $[2, 252] \times [2, 102]$. With no source and a large initial mobile pool, C_{10} decays exponentially to its equilibrium value C_{10}^{eq} . Since $C_{11} = 0$ throughout, $D_{xy} = 0$ and the problem reduces to a purely axial advection-diffusion; there is no cross-coupling to test. The blob drifts and diffuses slightly rightward during the fast transient and then freezes. Like Experiment 1, all snapshots after $t = 100$ are essentially identical.

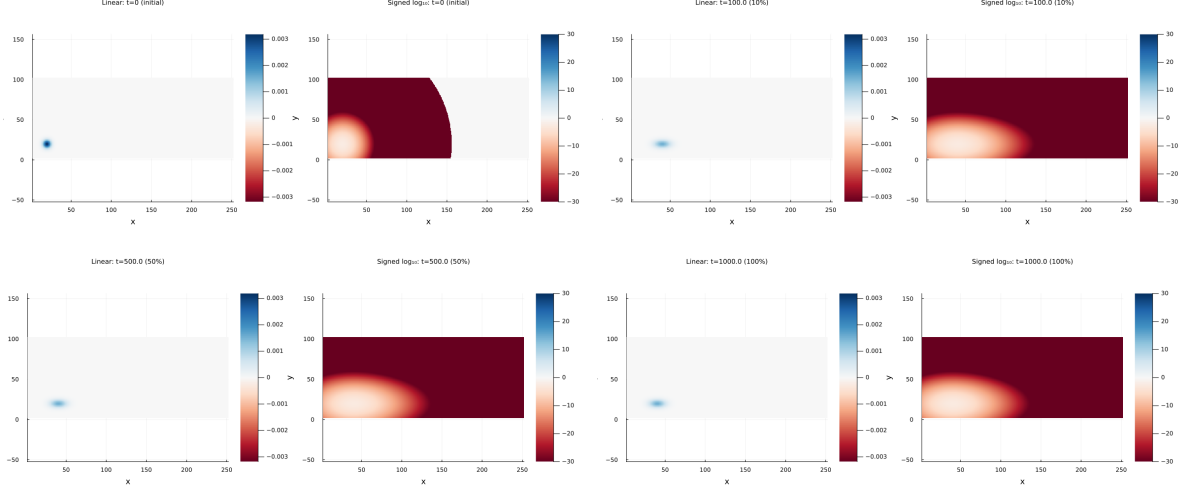


Figure 9: **Experiment 2 – High- C_{10} relaxation.** Snapshots at $t = 0, 100, 500, 1000$. The rightward diffusion visible between $t = 0$ and $t = 100$ is the only motion in the run; by $t = 100$ the system has frozen. Note the very broad but flat distribution: the large initial drift has spread the blob over a wide range before equilibration cuts off the advection.

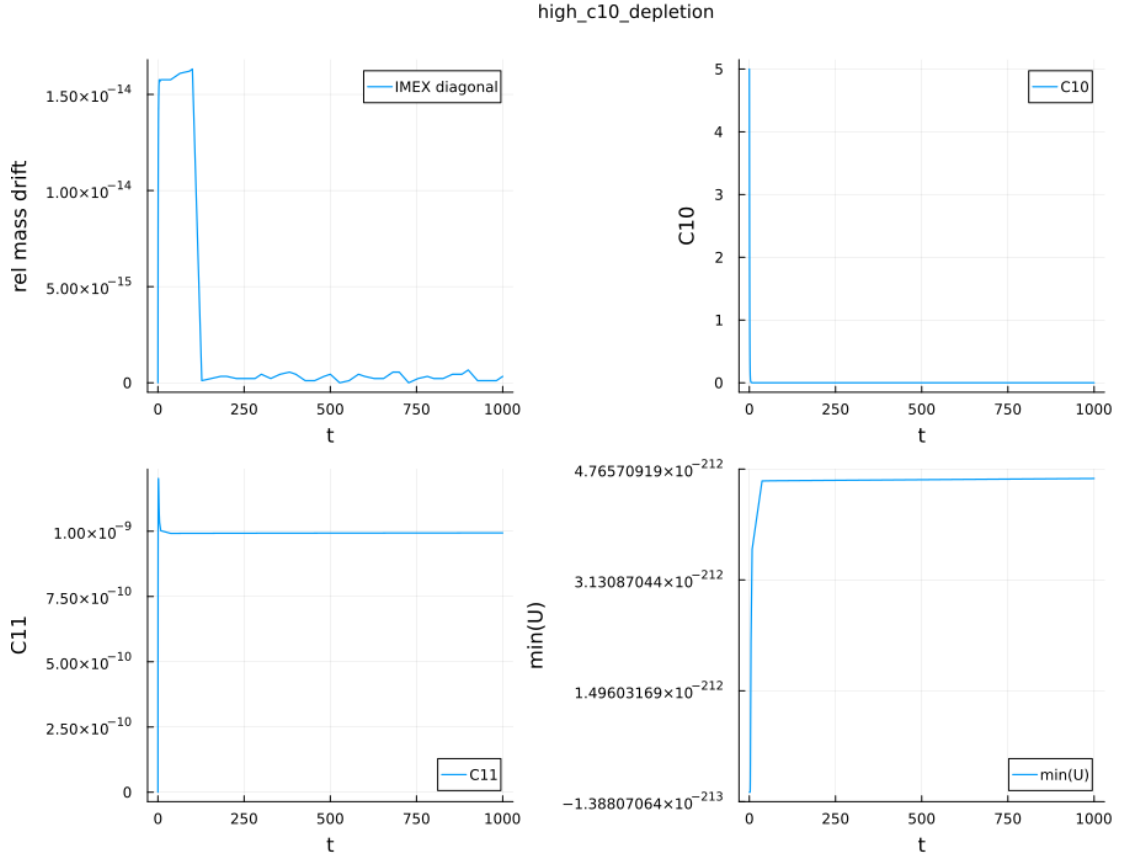


Figure 10: Diagnostics for Experiment 2. C_{10} drops from 5 to $\approx 2 \times 10^{-8}$ within the first checkpoint and plateaus; $C_{11} = 0$ throughout. $\min(U)$ stays at floating-point positive over the full horizon. The mass drift is negligible confirming that the single-step large- Δt integration is mass-conservative.

Experiment 3 – Cross-diffusion with non-square cells ($r = 1.2$)

$\varepsilon = 1.5$, $C_{10}(0) = C_{11}(0) = 1$, $G_{10} = 10^{-19}$, $G_{11} = 0$, $r = 1.2$ ($\Delta x = 0.6$, $\Delta y = 0.5$); Gaussian at $(5, 25)$, domain $[2, 302] \times [2, 102]$. The non-unit aspect ratio $r = 1.2$ exercises the full anisotropic diagonal stencil with distinct $\tilde{A}_x, \tilde{A}_y, \tilde{D}_2$. The initial mobile concentrations are high ($C_{10} = C_{11} = 1$) so the cross-diffusion is strong at $t = 0$; the NE–SW spreading is visible in the first checkpoint. After the fast equilibration both concentrations decay to near-zero and the system freezes, as in Experiments 1 and 2. The small residual source $G_{10} = 10^{-19}$ slowly tries to rebuild C_{10} over the 1time-horizon, but is too small to be visible in the plots.

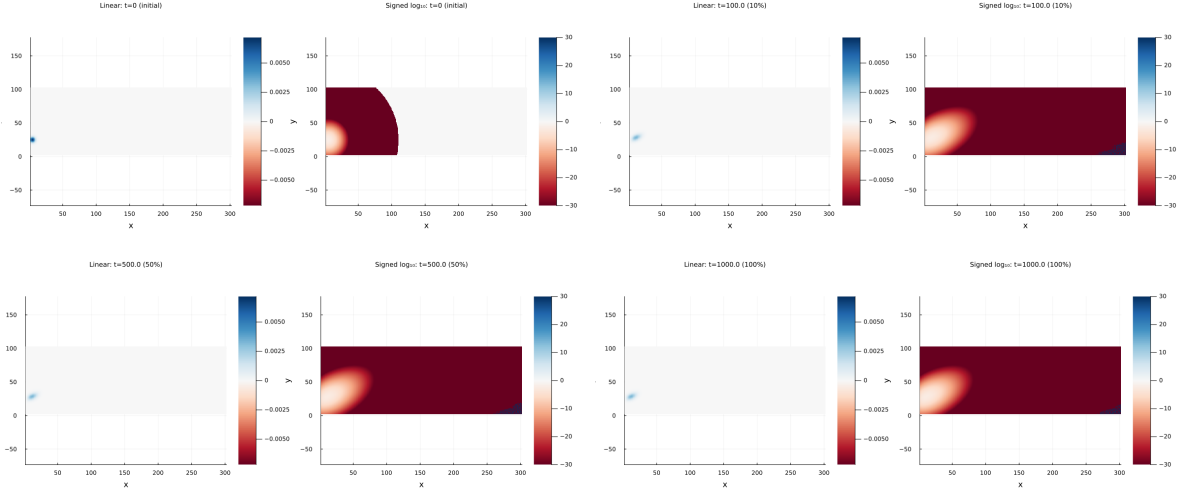


Figure 11: **Experiment 3 – Non-square cells ($r = 1.2$)**. The first checkpoint shows NE–SW elongation driven by $D_{xy} > 0$ at large $C_{11}(0) = 1$; subsequent snapshots are frozen. The elliptical shape reflects the anisotropic diagonal stencil for $r \neq 1$.

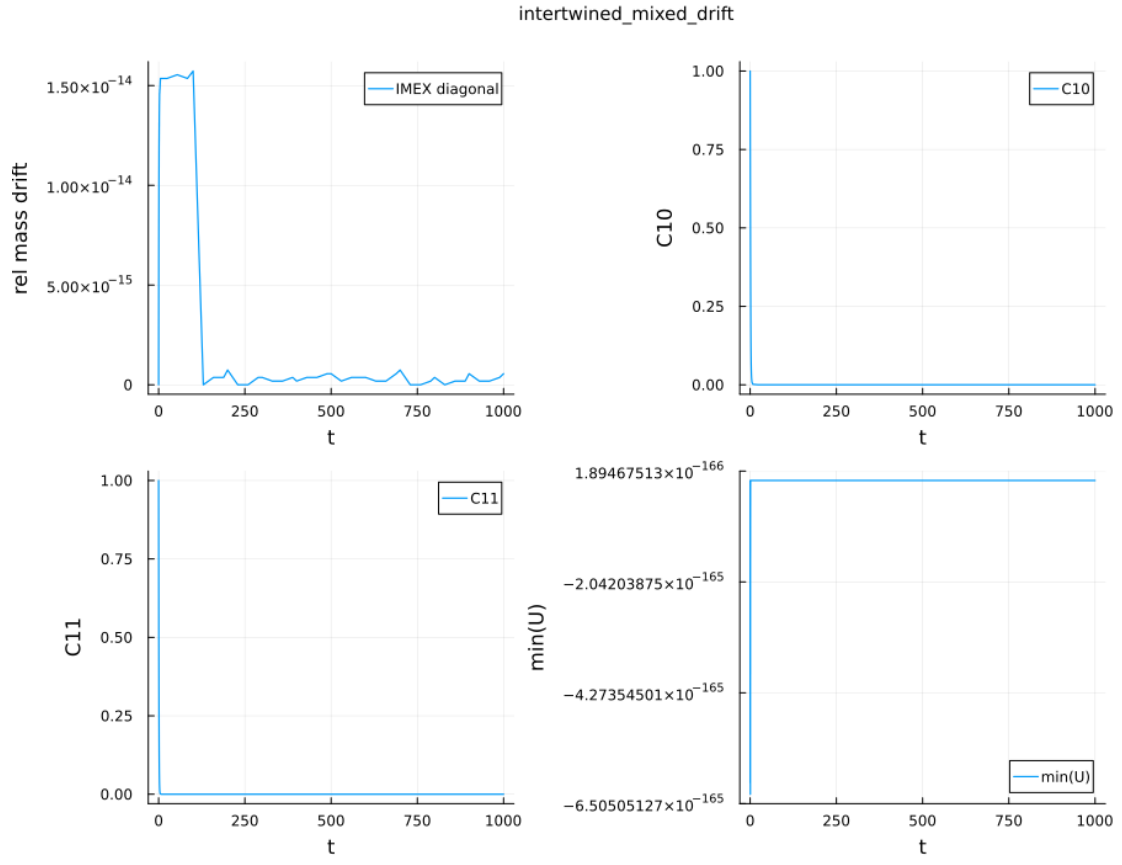


Figure 12: Diagnostics for Experiment 3.

Experiment 4 – Sustained drift with fixed mobile concentrations

$\varepsilon = 1$, $C_{10} = C_{11} = 0.01$ (fixed throughout), $G_{10} = G_{11} = 0$, $r = 1$; Gaussian at $(10, 10)$, domain $[2, 252] \times [2, 102]$. Fixing the mobile concentrations models a quasi-steady irradiation state where the drift $F_x \propto C_{10}\sqrt{x}$ is constant in time. This is the most demanding long-run test, as there is no equilibration freeze, so the advection solver is exercised for the full $T = 1000$ horizon and every checkpoint shows a distinct distribution.

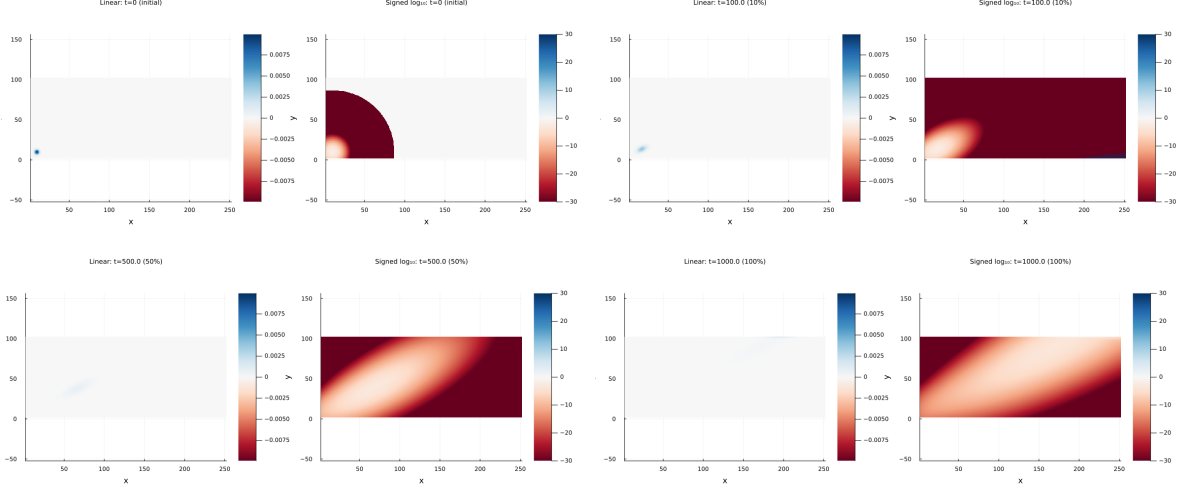


Figure 13: **Experiment 4 – Sustained drift (fixed mobile concentrations).** All four snapshots show a distinct distribution; the blob migrates to the right and upward and spreads diagonally under the constant $D_{xy} > 0$. By $T = 1000$ the peak has moved from $(10, 10)$ to approximately $(173, 82)$; the distribution is noticeably wider and lower-amplitude than at $t = 0$.

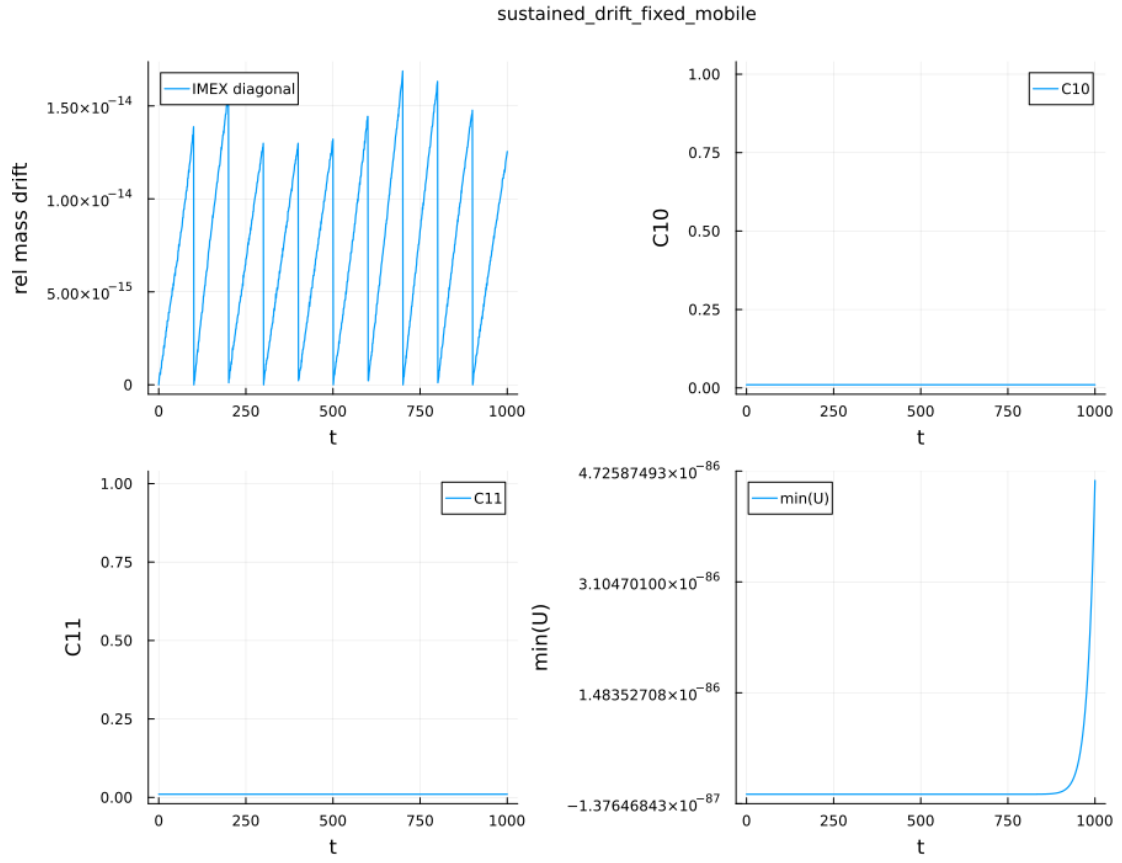


Figure 14: Diagnostics for Experiment 4.

Experiment 5 – Maximum physical $G_{11} = 10^{-17}$

$\varepsilon = 1$, $C_{10}(0) = 0.1$, $C_{11}(0) = 0.01$, $G_{10} = 0$, $G_{11} = 10^{-17}$, $r = 1$; Gaussian at $(10, 10)$, domain $[2, 252] \times [2, 102]$. The physical irradiation production rate for the second mobile species is bounded by $G_{11} \in [10^{-23}, 10^{-17}]$. The system behaves identically to the $G_{11} = 0$ case: after the fast equilibration both concentrations are negligible and the system freezes. All ten snapshots after $t = 100$ are indistinguishable from the $t = 100$ snapshot. This confirms that no physical G_{11} value can sustain cross-diffusion dynamics over long times. **This indicates that our simplified model maybe is not very interesting to investigate on long time-horizons for realistic parameters.**

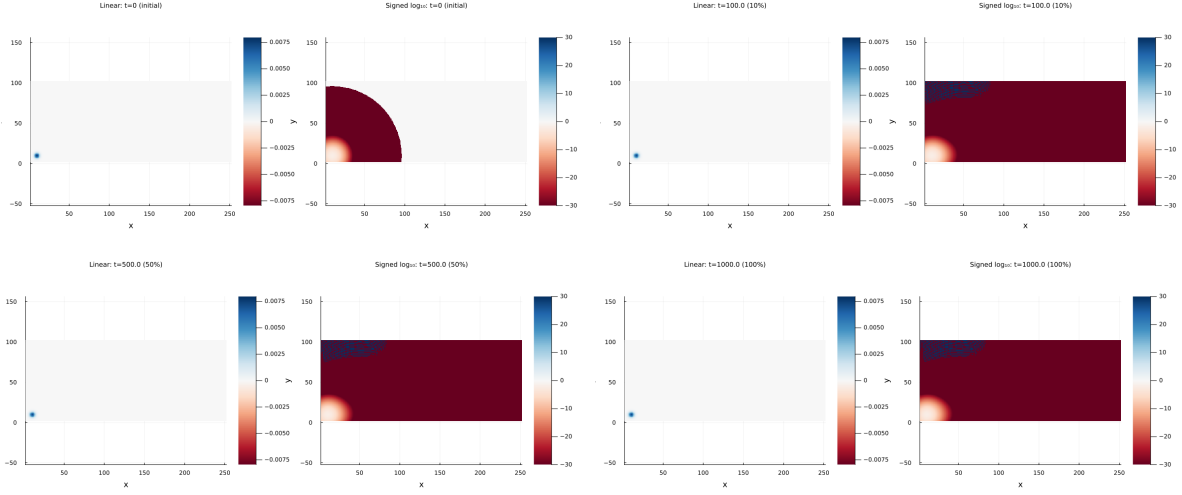


Figure 15: **Experiment 5 – Maximum physical $G_{11} = 10^{-17}$.** The snapshots after $t = 100$ are identical.

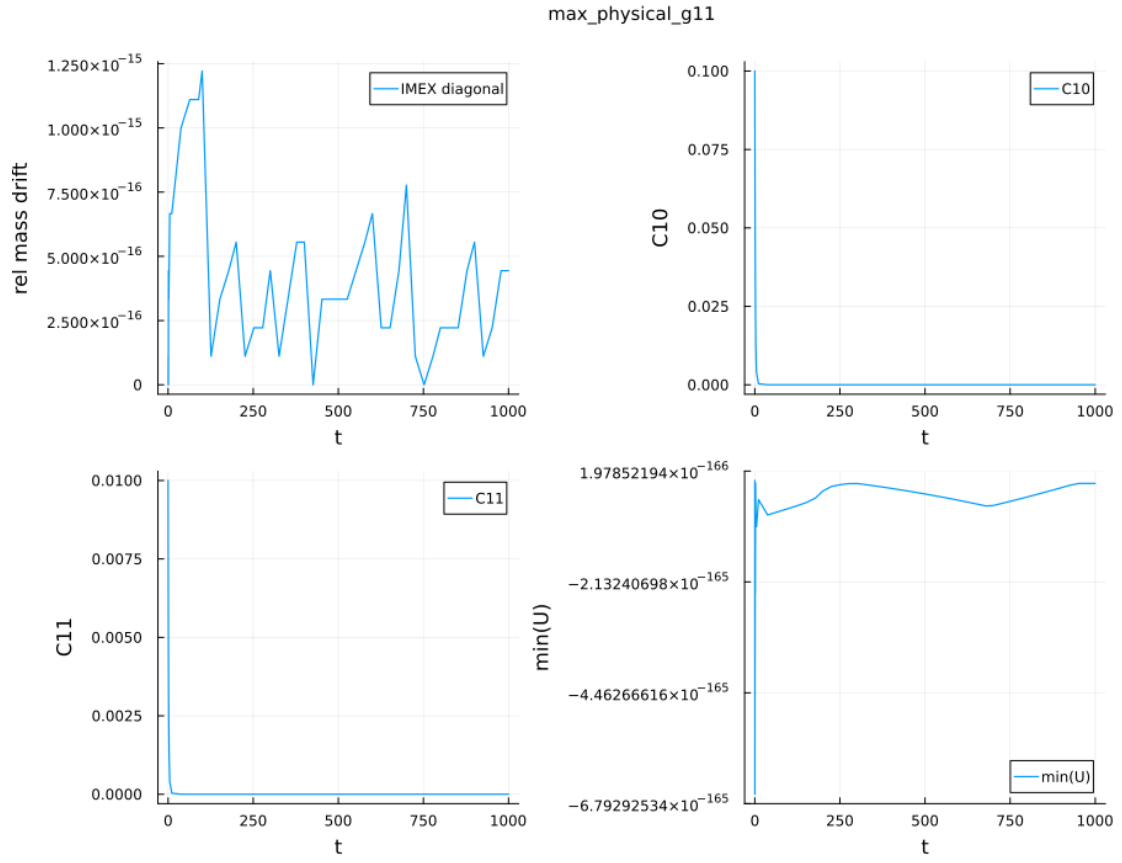


Figure 16: Diagnostics for Experiment 5.

Experiment 6 – Non-physical $G_{11} = 0.032$: illustration of sustained cross-diffusion

$\varepsilon = 1$, $C_{10}(0) = C_{11}(0) = 0$, $G_{10} = 0$, $G_{11} = 0.032$, $r = 1$; Gaussian at $(10, 10)$, domain $[2, 252] \times [2, 102]$. $G_{11} = 0.032$ is non-physical (bound is 10^{-17}) but chosen to illustrate the cross-diffusion regime. This value is chosen such that the concentrations of C_{01}, C_{11} keep growing slightly. All ten checkpoints show a continuously evolving distribution (not really visible here, but numerically the values kept changing).

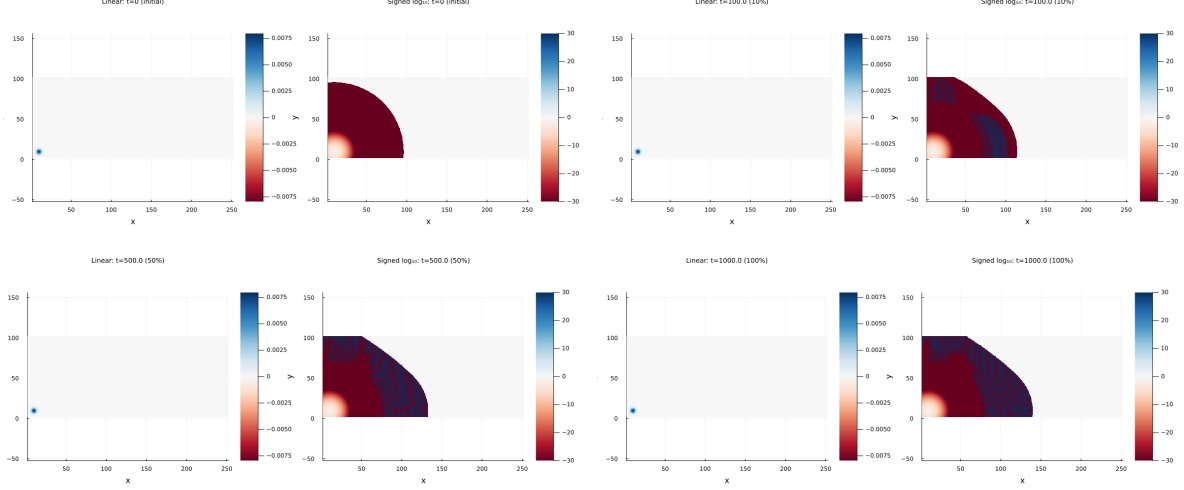


Figure 17: Experiment 6 – Non-physical $G_{11} = 0.032$, sustained cross-diffusion.

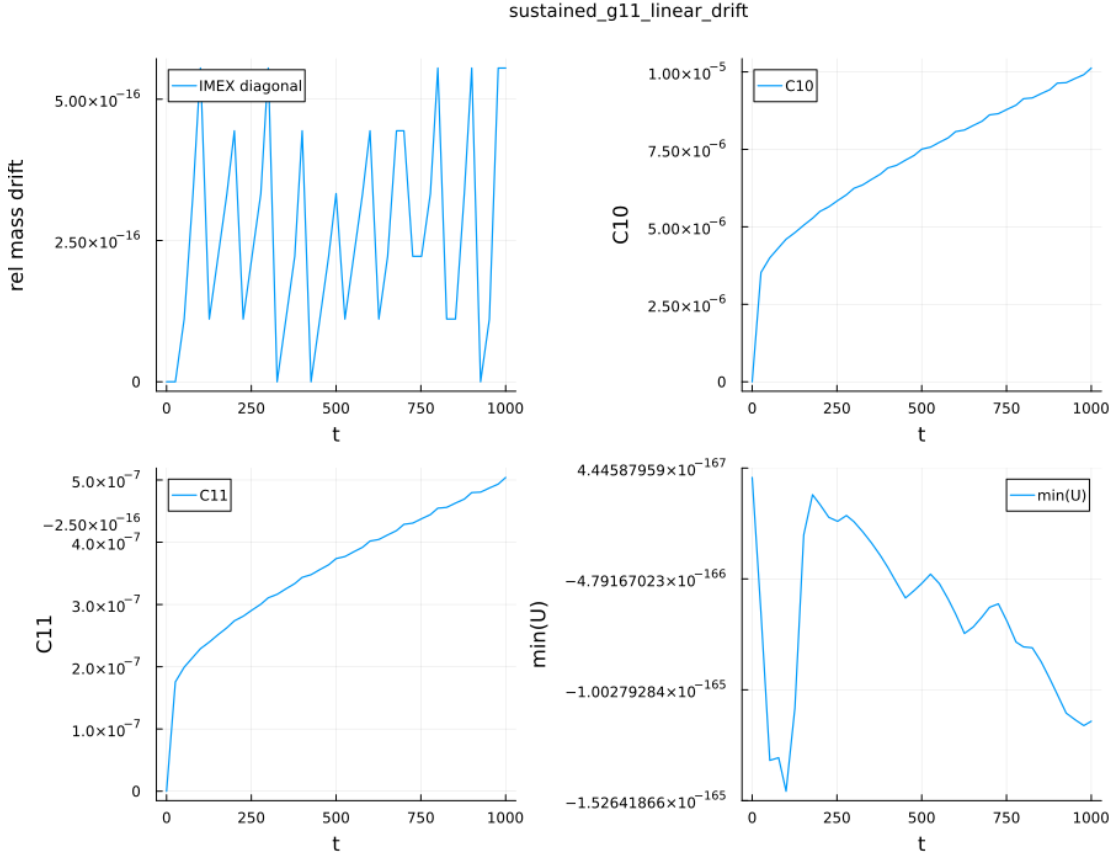


Figure 18: Diagnostics for Experiment 6.

9 Open todos

- Implement and validate the non-uniform geometric grid.
- Work on showing that the system mathematically also works in 3D. I tried this already, but failed because I was not really sure on what the assumptions are on such systems and how to generalize our diffusion method there. But this is probably a realistic goal to achieve in the next weeks.
- Continue finding an example where we have the cascade effect, turning negativity into an instability.
- Extend code to be able to use $D_{xy} > 0$ and $D_{xy} < 0$ automatically as outlined in this document.
- Kurganov comparison: implement the Kurganov advection flux and benchmark against MP5+RK3 in our scheme.